

Group Member Name	Student Number	Worked on Section?
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JOINT REPORT of Logan Braid and Oliver Wright and Luke Evans Unit name and Force – The Badgers Case against: John Doe	
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Summary

John Doe has had his hard drives seized to be investigated as he has been suspected of bird watching and being accused of bird abuse. So, as a team our goal is to find all evidence of any ornithological or illicit content of birds. After multiple different ways of searching the hard drive, our team gathered all the evidence we could find to make a report for a jury to decide on John Doe's verdict.

1. Description of Crime

In mid-June 2004, a Canon PowerShot SD100 camera was used to take a variety of photos, 19 of which were of an ornithological nature (*view figures 2.1a – 2.19a*), over the course of 2 weeks. The first bird related photo was taken at 18:28 on the 13th of June 2004 and the last was taken on 27th of June 2004 at 18:28.

On the 24th of January 2005, The Firefox web browser was used to install the Mozilla Thunderbird email client. From the 2nd of February at 14:11 through to 16:52, Firefox was used on John Doe's profile to browse the web, many of the sites visited included bird/birdwatching related media. On the 3rd of February the same profile was used to browse the web for more bird related media and information. Lastly, on the 9th of February the same profile was used to make the Firefox search "bird screensavers", the website "traveltex.com" was then visited with the specific webpages title being "Screensavers". All this activity can be seen in the Firefox History Timeline (*view appendix D*).

At 14:29 on 2nd of February 2005, the access of "nestboxtips.txt" was recorded (Figure 4.5.4 & 4.5.8) which is a text document containing lots of tips about making and keeping a nestbox.

At 12:22 on 3rd of February 2005, the access of a wav format audio file "aggressive_song.wav" (Figure 4.8.0-2) was recorded as coming from a connected removable drive, therefore the audio was unable to be found

At 15:02 on 3rd of February 2005, the access of a found htm document "ostbk2b2.htm" (Figure 4.6.5) was recorded which was found on the johndoe/MyDocuments section, it contains multiple references to chicks and birds. (Figure 4.4.24).

At 15:06 on 3rd of February 2005, the access of found image (Figure 2.46) was recorded which was found stored within bob, one of the non-administrator users that was found on the suspect drive (Figure 4.4.24).

At 15:52 on 3rd of February 2005, the access of a pdf file named "BirdingGuide.pdf" was recorded which was unable to be found. (Figure 4.7.0-1)

At 11:08 on the 9th of February 2005, 4 emails were received at John Doe's email address, jdoe@example.com, from person under the username ben@example.org. 3 of These emails had illicit ornithological images attached to them (*view figures 1.4, 1.5 & 1.7*) and one of them had a joke about eating a bird. A 5th email was received by jdoe@example.com at the same time as the other 4, but this time from the email mailinglist@birds.com. This email had a guide detailing information about birdwatching. One of the images sent by ben@example.org (*view figure 2.32a*) was taken using a Sony Cybershot camera (*view figure 2.32b*).

At 17:06 on the 9th of February 2005, an image file with the name "AlmondMarshGreatBlueHerronStalling.jpg" is recorded as being accessed (Figure 4.4.2 & 4.4.23). Though the image itself was not found when looking at the drive.

2. Description of Investigation

2.1 Job Description and Instructions

A digital forensics investigation has been carried out against John Doe as requested by Abertay University. We were asked to find any evidence relating to John Doe storing and looking at materials of an ornithological nature. The productions were found using several different techniques including searching for key phrases within the copy of the drive, searching for words related to birds, and searching common log files for anything which may be relevant. The found productions were then analysed and anything found that pointed to the existence of any other relevant files were then followed up on. Further details for productions were found using several software packages to give finer detail on the documents, giving information about how images were transferred and taken. These further productions help to give a more complete picture of what may have happened and what John Doe may have specifically done.

2.2 Description of Recovered / Examined Items

A physical hard drive (view figure 1.1) was recovered/ seized from John Doe's computer and imaged onto a Linux Ubuntu virtual machine. This allowed a full investigation of every file locally saved onto John Doe's Personal Computer. John Doe's computer was running on the windows XP operating system. The recovered evidence involves media files (images/gifs/audio recordings/documents), network files (history/ cookies) and email messages.

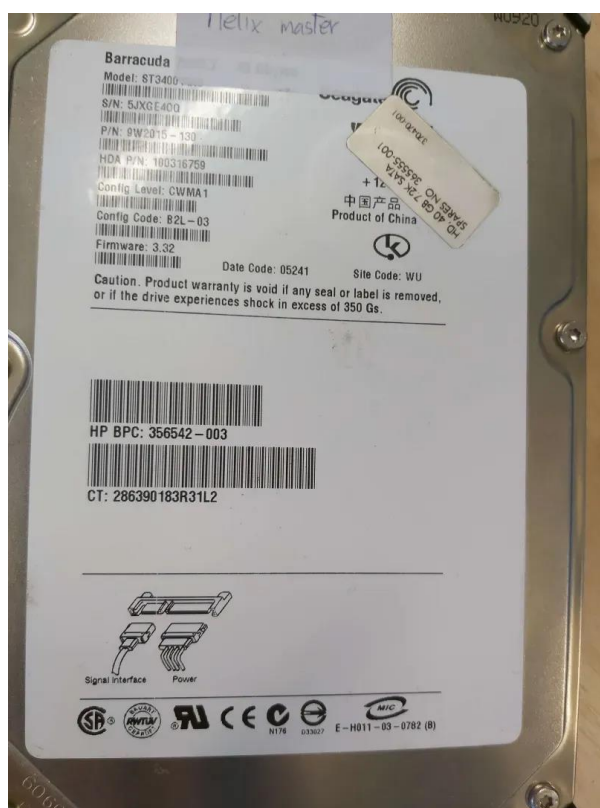


Figure 1.1: John Doe's seized Hard Drive

2.3 Methodology

Imaging Process

This is the process of copying the disk into a single file so that its contents can be safely explored. John Doe's seized hard drive was connected to the machine used to image the hard

drive using a SATA cable. A Linux tool, dcfldd, was used for the imaging process. On completion of the image, the hard drive's hash value was calculated using the md5sum command (view figure 1.2). This hash value was then compared to the originally captured hash value which ensures that no data has been created/modified/deleted during the imaging process. Md5sum was used throughout the entire investigation ensuring nothing changes on the hard drive during it.

Mounting Disk Process

This is the process of placing the disk's files into a readable state so its content can be explored without the possibility of making any changes to it. A loopback device is set up to access the disk image's files using an offset of 32256 bytes to skip over the partition table and meta data at the start of the image. The image is then mounted. The process of mounting John Doe's hard drive is used many times throughout the investigation therefore in future steps the methodology may talk about mounting the disk and that refers to this process.

Physical Searching

This process involves first using the copied disk drive to make it read-only so that nothing can be altered. To do that, we used the chmod command and just to make sure that nothing was changed, we used the md5sum command again. This showed that the hash value was the same so that we could continue. (See Figure 1.2 below)

D63dd1b8917ca28bac7c955fc3b6cd25

Figure 1.2: Hash Value for John Doe's Disk Drive

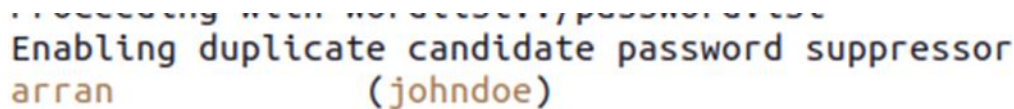
The next step is called File Carving, where we use a command called foremost to make a file into a jpg file so that we can see all the images John Doe has on his hard drive in his folders. Once we had all his jpg images, we had to sift through all of them and find every instance of anything ornithological or anything illicit of birds, including some images of a man dressed as a bird. (See "All images from jpg file relating to birds" in Appendix A)

Once we looked through all the images, we took his copied hard drive and put it into a software called Autopsy. Autopsy allows us to look more thoroughly through his hard drive and can let us recover anything that could have been previously deleted.

Password Cracking

This process was needed as we found an encrypted folder called birdpics.gpg on John Doe's hard drive. Due to the name of it, we had to access it as it could be relevant to the case because, as the name says, there could be bird pictures in this folder. So, the first step we did was to figure out his password, and there are multiple ways to do this. There is the brute force method and the dictionary attack method, or we could look through his hard drive for an application called Thunderbird and see if his passwords are on there. We did both the brute force method and the dictionary attack to see if we could find the password or multiple. Before we do anything though, we had to use a loopback mounting technique. This is to access his hard drive and make a copy of the birdpics.gpg and a directory called GnuPG, which is full of folders. We believe we can find his password through these protected folders because these folders are protected by a weaker password that we can take advantage of by using a brute force method.

First was the brute attack. We used a program called John the Ripper. This program does brute force and dictionary attacks, so we will use it for both attacks. We attacked a folder found in the GnuPG directory called secring.gpg, as we believe the key/password is in here. So, after the attack we discovered that John Doe's password is 'arran'. (See Figure 1.3 for the part where John the Ripper discovered John Doe's password). To check it worked, we imported it into the secring.gpg folder and see if it would work, and it did.



```
Enabling duplicate candidate password suppressor
arran (johndoe)
```

Figure 1.3: John the Ripper finding John Doe's password

Logical Searching

Whitelisting

A whitelist was used to look for all files that had been created/downloaded onto the suspect hard drive (e.g. non-base operating system files). To do this md5deep was used to retrieve the hash value and file pathway for every single file on the suspect hard drive and save it into a text file named "johnDoe.nameAndmd5List.txt". This was also done on a clean Windows XP image saving the same details into a text file named "winXP.nameAndmd5List.txt". Using these two text files, grep was used to only show strings in "johnDoe.nameAndmd5List.txt" that are not in "winXP.nameAndmd5List.txt" hence filtering out all the base Windows XP operating system files. The results of this were saved into a text file named "whitelistOutput1.txt".

Now that the whitelist has been made, grep was used once again to make 3 separate text files from "whitelistOutput1.txt". The first file searched "whitelistOutput1.txt" for bmp files and output the file hashes and pathways into a new txt file, "whitelistOutputBMP.txt". The second searched for jpg files, and output the results to "whitelistOutputJPG.txt", and the third searched for wav files and output the results to "whitelistOutputWAV.txt".

Directed Searching

The image of John Doe's personal computer which was created in the original imaging process was verified against the hash value as being the same. Then the image was mounted onto the analysis machine and the registry files were collected from the drive. These were found within system32/config (SAM, SECURITY, software, and system) as well as another in the johndoe user's folder (NTUSER.DAT), these files were then copied out into a separate folder on the analysis machine to be used in the following analyses.

User Information

The chntpw command was used on the SAM (secure access management) file in order to ascertain the users and permissions of the users on the personal computer, this showed that the default administrator account and john doe had the elevated administrative permissions while the other users all didn't, including bob and jane the two other non-default users. (See Figure 4.1)\=

Software & Computer Information

The software section of the HKLM registry was opened and within that "App Paths" was traversed to in FRED this is found at "software\Microsoft\Windows\CurrentVersion\App

Paths” within it the found software packages were looked over, however only two of note were found (See Analysis) (Figure 4.2.0).

Next the system section of the HKLM was opened and “MountedDevices” within it was checked, this listed 5 devices. (See figure 4.2.1-2 & 4.2.4). The “Controlset001\Enum\USBSTOR” section under the system hive in the HKLM was then opened and checked within it the files for one USB connected drive existed. This shows that only one USB storage drive has been recorded as being connected at any point on the computer. (Figure 4.2.3)

The “ComputerDescriptions” section of NTUSER.DAT was checked, found under “Software\Microsoft\Windows\CurrentVersion\Explorer\ComputerDescriptions”, it showed several connections to Samba 2.2.8a and 3.0.10Debian (versions of a software often used to connect Linux & Unix systems to Windows). Aswell as several connections to other XP computers and a laptop (Figure 4.3). A single connection to a “Fabio’s new Sony” exists likely being the same Fabio with a laptop connection and another PC connection with no value stored. This Sony however likely is unimportant to the case due to the found images being tagged as taken with a Canon.

MRU Data

Within the NTUSERS.DAT registry file the MRU (most recently used) file data was looked at in depth starting with the “RecentDocs” section. In this section 10 file names were found. The list “MRUListEx” with each number relating to the value of the key shows the order of recency (see Figure 4.4.0 - 4.4.10) and as such the images are shown in the appendix in the same order. The found file names were then followed up on by searching the image’s standard filesystem for the names using the find command. The images were then retrieved from the file system which all showed bird imagery and are supplied in the appendix in order from most recent (see Figure 2.46-52). The OpenSaveMRU was then checked and roughly corroborated the most recent file names from before, showing the same order apart from not having the three documents that were unfindable from “RecentDocs” (See Figure 2.53-55). The OpenSaveMRU jpg section like the “RecentDocs” jpg section contained 10 files and instead of the three missing files it had 3 older files that weren’t contained in “RecentDocs”. Finally, the last time that the most recent file was accessed was found through going to the jpg section of the “RecentDocs” using the Autopsy software.

Next the MRU was again checked for other file types following a similar pattern of searching “RecentDocs” for each file type, doing a find for each file listed. Afterwards the OpenSaveMRU was once again checked to corroborate, and the time stamps found. This was done for (in order) txt, htm, pdf, and wav file types.

Network Searching

Emails

Autopsy was used to recovered emails retrieved by the email address “jdoe@example.com” and sent from the email address “ben@example.org”.

Firefox

The Mozilla Firefox History file “history.dat” was located at “Documents and Settings/johndoe/Application Data/Mozilla/Firefox/Profiles/w4nf3obl.default”. The form history file “formhistory.dat” and cookies file “cookies.txt” were also found at the same place. A tool called “MZHistoryView” was used to parse both the history files into a readable format. The MZHistoryView inbuilt function, to create a HTML report, was then used to generate a more readable report (*view Appendix C for the full HTML report*). MZHistoryView

only works on windows machines so before using the tool, the Firefox files were transferred over to a windows machine via a shared network folder.

Manual Searching

Some evidence was also found on John Doe's hard drive by simply mounting the disk and manually looking through the file systems (*view figures: 2.33 – 2.46 in Appendix A*).

2.4 Analysis

Manually Found Evidence

This section refers to the "birdwatching.doc" document that was found on the hard drive (*view figures 3.9 - 3.18*). The document lists a copious amount of information about birds specifically in Thailand. It has information on the how to identify different species of birds. It also details the best times to see birds and where they are likely to be seen. It then lists which specific species of birds are spotted in certain environments. Furthermore, the document also provides "birdwatching tips" which gives advice on how one could view birds more closely without them noticing.

In the "My Documents" folder on John Doe's windows profile, a file named "nestboxtips.txt" was found (*view figure 3.19*). This text file contains information on how to upkeep a bird nestbox as well as details on how to make the birds as comfortable as possible.

Another document named "Doc1.doc" was found on the hard drive (*view figure 3.4*). This document contained a very zoomed in image where the only visible elements of it appeared to be a branch and some feathers. The full image was recovered and found to be another bird image (*view figure 2.56*).

Firefoxwallpaper.bmp is a picture of a bird (*view figure 2.57*) found in john doe's Firefox folder, name and location implies it was used/ to be used as his Firefox browsers wallpaper.

Metacam and Nano findings

We used Metacam and nano commands to go through all the images and see if any of the images were different. We found many images taken on a Canon PowerShot SD100, then narrowed it down to 19 images on the camera that can be used as evidence as they had images of birds or bird-related items in the images. We also took the information of the Canon PowerShot SD100 as evidence of when each photo was taken to create our timeline. (See Figure 2.1 to Figure 2.19 in Appendix A to see all the images and information retrieved.) There was also one image of a bird we found taken on a Canon EOS-1DS and this image was taken a year before all the other images were taken. (See Figure 2.20 in Appendix A for the image and the information retrieved.)

Whitelists

While many of the jpg and bmp files were found by autopsy and other previously mentioned methods, the whitelist did prove to be useful for double checking that as much evidence as possible was found. The "whitelistOutputWAV.txt" proved to be incredibly useful as well as it

found a file named “aggressive_song.wav”. The audio in the WAV file appeared to be of bird whistling/ chirping noises, however, the species of bird is unidentified.

Password Cracking Findings

After discovering the password to be “arran”, we tried using it to gain access to the birdpics.gpg folder, and sure enough we got in. We found that the folder was a zip folder full of images, so we unzipped the folder and found five images of birds. (See Figures 2.21 to Figure 2.25 in Appendix A for all the images of the birds.)

Once we had done the brute force method, we tried the dictionary attack on the secring.gpg to see if it gave us the same result or different results, so we used all the words in John Doe's hard drive for the dictionary attack to see if we can get a match. After this test, we found the same password 'arran' as the brute force method, but the dictionary method found it faster. So, we know both attacks work correctly. After doing both tests, we must unmount the loopback mounting technique so that it doesn't cause any problems with the drive and ensure that nothing has been altered.

Software & Computer Information Analysis

References to the two executables “firefox.exe” and “thunderbird.exe” were found within the HKLM app paths section. Firefox is a standard browser that further information was found in within other sections, and thunderbird is an email client which was used for emails which are found and analysed within other sections as well.

The existence of the 5 past connected drives was looked through. These devices being a floppy disk drive and cd drive that weren't of interest the standard C drive, an E drive that was the connection for Removable Media, and an F drive that had little data about it just having standard partition details. (Figure 4.2.0).

The one USB storage drive connected found in USBSTOR was recorded with the last access time that was recorded about the USB being on 09/02/2005 at 03:10:17, with a time of original connection going back to 02/02/2005 at 16:29:57 or before. Based on recorded times within the file (Figure 4.2.3).

MRU Data

Of the 10 file names that were searched for using the find command following the searching through of the jpg section of RecentDocs 7 were found (see Figure 4.4.11). All of which contained bird imagery (Figure 2.46-52). A major thing of note was the finding of the image “ready2fledge.jpg” (Figure 2.46) instead of under the files of user “johndoe” being under the user “bob”. Within Autopsy the last accessed time was ascertained using the “Modification Time:” section of the greater section, which shows that “AlmondMarshGreatBlueHeronStalling.jpg” was last accessed at 17:06:28 GMT on 09/02/2005(see Figure 4.4.23). Though that image was not one that could be found on the drive.

The files found in the OpenSaveMRU also contained bird related imagery (see Figure 2.53-55). The OpenSaveMRU timestamp was also checked in the same section showing “ready2fledge.jpg” was last accessed at 15:06:42 GMT on 03/02/2005(see Figure 4.4.24).

Following the pattern established in methodology, Two references to txt documents were found in the “RecentDocs” area (Figure 4.5.0-2), cookies.txt and nestboxtips.txt with cookies.txt (Figure 3.2-3) being the most recent being accessed last at 12:19:07 on 03/02/2005 (based on just RecentDocs see Figure 4.5.7), and nestboxtips.txt being accessed being corroborated by both but with a last access only stated on OpenSaveMRU(Figure 4.5.8) accessed last at 14:29:30 on 02/02/2005.

References to two htm documents were found in the “RecentDocs” and “OpenSaveMRU” area (Figure 4.6.0-2), with the most recently opened being ostbk2b2.htm at 15:02:45 on 03/02/2005 (based on both see Figure 4.6.3-4) the htm documents both contain references to birding.

A single reference to a “birdingGuide.pdf” being accessed was found in “RecentDocs” no corroboration with the “OpenSaveMRU” due to the nature of it being a pdf. (Figure 4.7.0-1) it was accessed at 15:52:01 on 03/02/2005.

A reference to a wav audio file was found in the “RecentDocs” and “OpenSaveMRU” area (Figure 4.8.0-1), aggressive_song.wav at 12:22:51 on 03/02/2005 (based on both see Figure 4.8.2-4)

Network Analysis

In John Doe’s Firefox cookies.txt file (*view figures 3.2 – 3.3*), it showed the website “www.haiths.com” which is a birdwatching website. The Haiths website provides bird related products like bird food, bird feeders and bird nest boxes. The cookies also show that a session was started on the website. This could mean that it is John Doe’s first visit which makes sense as he then navigates to the “about” page. However, sessions also start when a user logs in, so it could mean that John Doe is a frequent user who has an account on the bird watching website.

The Firefox “history.dat” file (*view figures 3.5 – 3.8*) contained extensive material of avian nature. The types of webpages visited include bird related articles, bird related books, bird wallpapers, bird images, birdwatching shops, bird indexes, bird feeder building plans, bird house building plans, birdwatching guides, bird movies, bird mating calls and bird screensavers. All the bird related searches found in the “history.dat” file can be seen in the Firefox History Timeline (*view appendix D*).

Emails

The following email evidence shows an implied dialogue between John Doe using the email address “jdoe@example.com” with someone referred to as “ben” using the email address “ben@example.org”. The first email (*view figure 1.4*) from ben has a subject of “good pics”, and the email reads “Hi thought you’d like these enjoy”. Attached to this email are 3 images of birds (*view figures 2.26 – 2.28*).

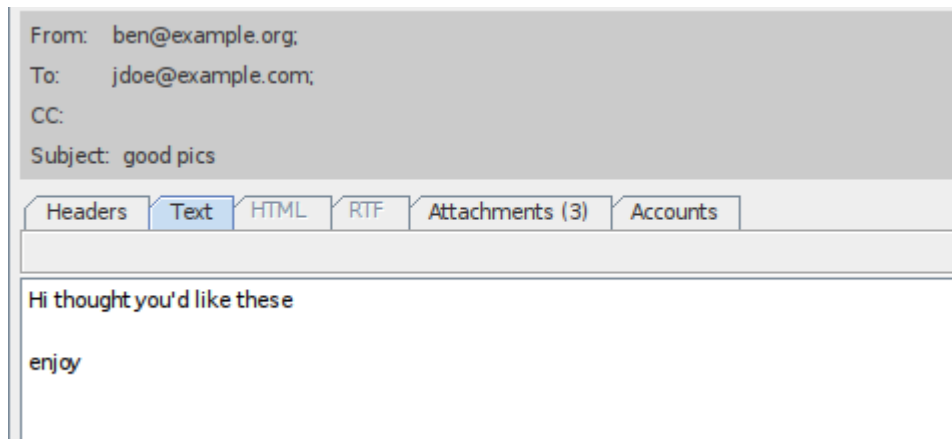


Figure 1.4: Emails received by jdoe@example.com (1/5)

The next email (*view figure 1.5*) from ben has the subject “some more good ones” and has the body “Thanks for the pics you sent me here are some I really like”. This email implies that John Doe sent pics back to ben and context leads to believe that they may have been bird pics. This email also has 5 more bird images attached (*view figures 2.28 – 2.32*). The last image (*view figure 2.32*) was taken using a Sony Cybershot camera on the 30th of December 2002.

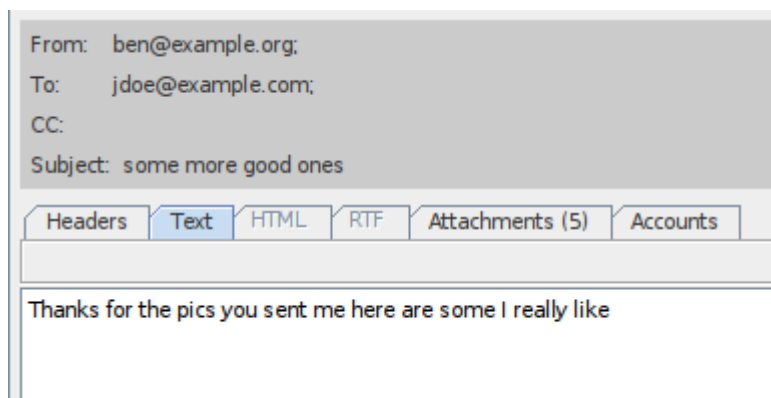


Figure 1.5: Emails received by jdoe@example.com (2/5)

The third email (*view figure 1.6*) sent from ben has the subject “expensive birds” and consists of a story, potentially a joke, about a cockatoo being bought and eaten.

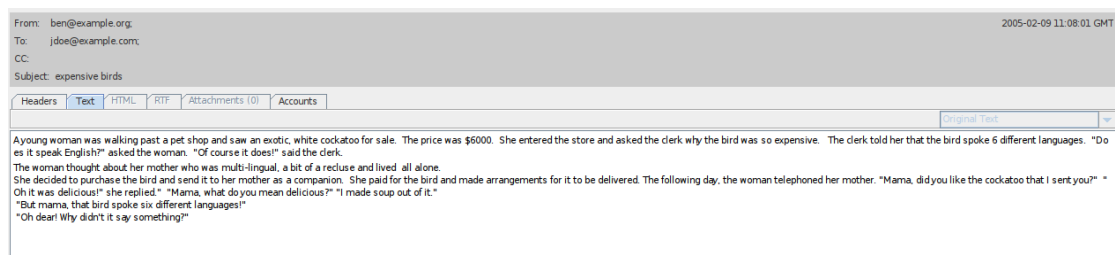


Figure 1.6: Emails received by jdoe@example.com (3/5)

The last email from ben (*view figure 1.7*) has the exact same subject and message as the first one. It only has one attached image (*view figure 2.27*) which is the same as one of the three images previously attached in ben's first email.

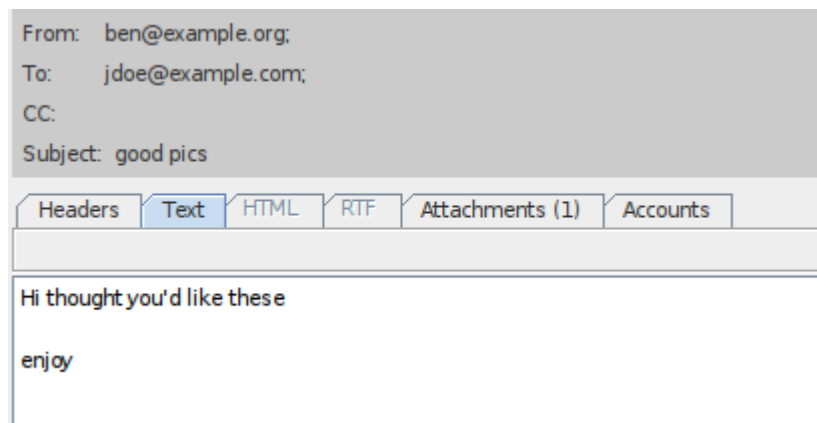


Figure 1.7: Emails received by jdoe@example.com (4/5)

John Doe also received an email from mailinglist@birds.example.com (*view figure 1.8*), with the subject “How to Identify Birds”. The email message then proceeds to detail different tips and techniques about how to identify different types of birds. This email address implies that John Doe’s email was voluntarily signed up for this newsletter to receive information/advice on different bird watching techniques.

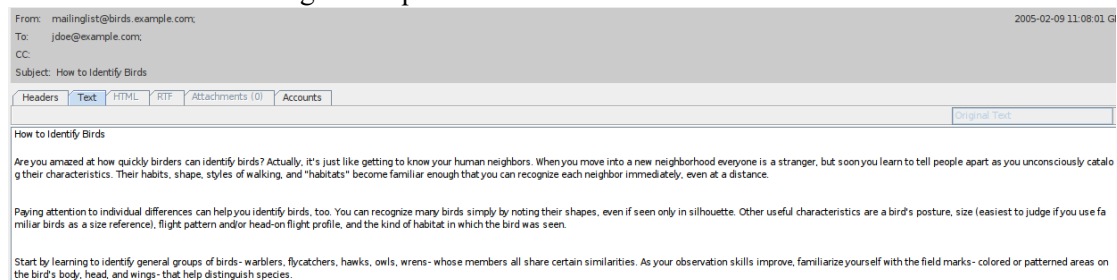


Figure 1.8: Emails received by jdoe@example.com (5/5)

3. Conclusion

The investigation concluded having found an overwhelming amount of evidence of ornithological-related media, including images, audio files, GIFS, and informative documents involving birds or bird-adjacent themes. Evidence found over a variety of different cameras found multiple images of birds and images of items with ornithological nature. With a quick manual search through the hard drive, we were able to find plenty more images of birds just in the folders. More evidence was also found, such as email conversations about birds, bird-related internet searches and informative guides about, or including, tips and techniques on birdwatching, this includes information such as, locating birds, identifying birds, how to not startle them and strategies of getting closer to them.

In conclusion, there is a substantial amount of evidence that suggests that John Doe’s Hard drive and email were used for acquiring, possessing, and distributing illicit material of an ornithological nature.

4. Equipment Required for Court Proceedings

Evidence:

- John Doe’s Hard Drive

Software/Tools used:

- Ubuntu Virtual machine
- Autopsy
- John the Ripper
- Linux Commands
- GPG
- MZHistoryView
- FRED

Appendices

Appendix A – Media Evidence

Images Taken from “Canon PowerShot SD100” Camera



Figure 2.1a: 01243351.jpg.

```
File: 01243351.jpg
Standard Fields -----
Image Creation Date: 2004:07:02 17:27:53
Make: Canon
Model: Canon PowerShot SD100
Software Version: Adobe Photoshop Elements 2.0
X Resolution: 180 Pixels/Inch
Y Resolution: 180 Pixels/Inch
YCbCr Positioning: Center of Array
EXIF Fields -----
Image Capture Date: 2004:06:27 18:28:34
Image Digitized Date: 2004:06:27 18:28:34
```

Figure 2.1b: Figure 1a's image information.



Figure 2.2a: 02791503.jpg.

```

File: 02791503.jpg
Standard Fields -----
Image Creation Date: 2004:07:02 17:30:41
Make: Canon
Model: Canon PowerShot SD100
Software Version: Adobe Photoshop Elements 2.0
X Resolution: 180 Pixels/Inch
Y Resolution: 180 Pixels/Inch
YCbCr Positioning: Center of Array
EXIF Fields -----
Image Capture Date: 2004:06:13 19:44:11
Image Digitized Date: 2004:06:13 19:44:11

```

Figure 2.2b: Figure 2a's image information.



Figure 2.3a: 02792151.jpg.

```

File: 02792151.jpg
Standard Fields -----
Image Creation Date: 2004:07:02 17:27:53
Make: Canon
Model: Canon PowerShot SD100
Software Version: Adobe Photoshop Elements 2.0
X Resolution: 180 Pixels/Inch
Y Resolution: 180 Pixels/Inch
YCbCr Positioning: Center of Array
EXIF Fields -----
Image Capture Date: 2004:06:27 18:28:34
Image Digitized Date: 2004:06:27 18:28:34

```

Figure 2.3b: Figure 3a's image information.

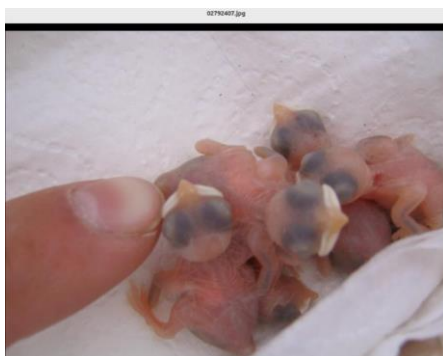


Figure 2.4a: 02792407.jpg.

```

File: 02792407.jpg
Standard Fields -----
Image Creation Date: 2004:07:02 17:27:53
Make: Canon
Model: Canon PowerShot SD100
Software Version: Adobe Photoshop Elements 2.0
X Resolution: 180 Pixels/Inch
Y Resolution: 180 Pixels/Inch
YCbCr Positioning: Center of Array
EXIF Fields -----
Image Capture Date: 2004:06:27 18:28:34
Image Digitized Date: 2004:06:27 18:28:34

```

Figure 2.4b: Figure 4a's image information.



Figure 2.5a: 02903551.jpg.

```
File: 02903551.jpg
Standard Fields -----
Image Creation Date: 2004:07:02 17:47:26
Make: Canon
Model: Canon PowerShot SD100
Software Version: Adobe Photoshop Elements 2.0
X Resolution: 180 Pixels/Inch
Y Resolution: 180 Pixels/Inch
YCbCr Positioning: Center of Array
EXIF Fields -----
Image Capture Date: 2004:06:18 23:33:33
Image Digitized Date: 2004:06:18 23:33:33
```

Figure 2.5b: Figure 5a's image information.



Figure 2.6a: 02962639.jpg.

```
File: 02962639.jpg
Standard Fields -----
Image Creation Date: 2004:07:02 17:49:33
Make: Canon
Model: Canon PowerShot SD100
Software Version: Adobe Photoshop Elements 2.0
X Resolution: 180 Pixels/Inch
Y Resolution: 180 Pixels/Inch
YCbCr Positioning: Center of Array
EXIF Fields -----
Image Capture Date: 2004:06:17 18:43:03
Image Digitized Date: 2004:06:17 18:43:03
```

Figure 2.6b: Figure 6a's image information.



Figure 2.7a: 02963839.jpg.

```

File: 02963839.jpg
Standard Fields -----
Image Creation Date: 2004:07:02 17:46:10
Make: Canon
Model: Canon PowerShot SD100
Software Version: Adobe Photoshop Elements 2.0
X Resolution: 180 Pixels/Inch
Y Resolution: 180 Pixels/Inch
YCbCr Positioning: Center of Array
EXIF Fields -----
Image Capture Date: 2004:06:17 18:42:34
Image Digitized Date: 2004:06:17 18:42:34

```

Figure 2.7b: Figure 7a's image information.



Figure 2.8a: 03074343.jpg.

```

File: 03074343.jpg
Standard Fields -----
Image Creation Date: 2004:07:02 17:46:36
Make: Canon
Model: Canon PowerShot SD100
Software Version: Adobe Photoshop Elements 2.0
X Resolution: 180 Pixels/Inch
Y Resolution: 180 Pixels/Inch
YCbCr Positioning: Center of Array
EXIF Fields -----
Image Capture Date: 2004:06:18 23:31:50
Image Digitized Date: 2004:06:18 23:31:50

```

Figure 2.8b: Figure 8a's image information.

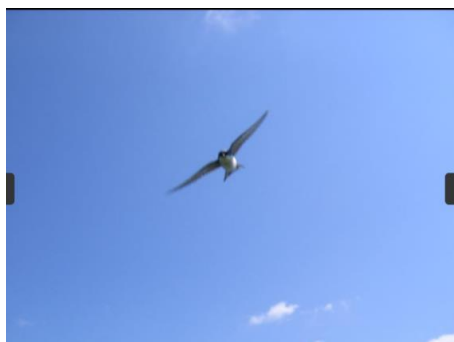


Figure 2.9a: 03112503.jpg.

```

File: 03112503.jpg
Standard Fields -----
Image Creation Date: 2004:07:02 17:41:45
Make: Canon
Model: Canon PowerShot SD100
Software Version: Adobe Photoshop Elements 2.0
X Resolution: 180 Pixels/Inch
Y Resolution: 180 Pixels/Inch
YCbCr Positioning: Center of Array
EXIF Fields -----
Image Capture Date: 2004:06:22 18:56:18
Image Digitized Date: 2004:06:22 18:56:18

```

Figure 2.9b: Figure 9a's image information.



Figure 2.10a: 03180791.jpg.

```
File: 03180791.jpg
Standard Fields -----
Image Creation Date: 2004:07:02 17:49:50
Make: Canon
Model: Canon PowerShot SD100
Software Version: Adobe Photoshop Elements 2.0
X Resolution: 180 Pixels/Inch
Y Resolution: 180 Pixels/Inch
YCbCr Positioning: Center of Array
EXIF Fields -----
Image Capture Date: 2004:06:18 23:31:56
Image Digitized Date: 2004:06:18 23:31:56
```

Figure 2.10b: Figure 10a's image information.

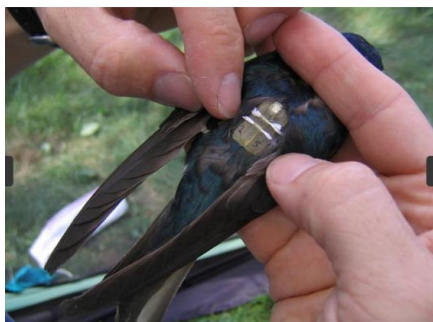


Figure 2.11a: 03181927.jpg.

```
File: 03181927.jpg
Standard Fields -----
Image Creation Date: 2004:07:02 17:50:06
Make: Canon
Model: Canon PowerShot SD100
Software Version: Adobe Photoshop Elements 2.0
X Resolution: 180 Pixels/Inch
Y Resolution: 180 Pixels/Inch
YCbCr Positioning: Center of Array
EXIF Fields -----
Image Capture Date: 2004:06:18 23:32:28
Image Digitized Date: 2004:06:18 23:32:28
```

Figure 2.11b: Figure 11a's image information.



Figure 2.12a: 03184607.jpg.

```

File: 03184607.jpg
Standard Fields -----
Image Creation Date: 2004:07:02 17:46:10
Make: Canon
Model: Canon PowerShot SD100
Software Version: Adobe Photoshop Elements 2.0
X Resolution: 180 Pixels/Inch
Y Resolution: 180 Pixels/Inch
YCbCr Positioning: Center of Array
EXIF Fields -----
Image Capture Date: 2004:06:17 18:42:34
Image Digitized Date: 2004:06:17 18:42:34

```

Figure 2.12b: Figure 12a's image information.



Figure 2.13a: 03185759.jpg.

```

File: 03185759.jpg
Standard Fields -----
Image Creation Date: 2004:07:02 17:27:29
Make: Canon
Model: Canon PowerShot SD100
Software Version: Adobe Photoshop Elements 2.0
X Resolution: 180 Pixels/Inch
Y Resolution: 180 Pixels/Inch
YCbCr Positioning: Center of Array
EXIF Fields -----
Image Capture Date: 2004:06:13 20:39:11
Image Digitized Date: 2004:06:13 20:39:11

```

Figure 2.13b: Figure 13a's image information.



Figure 2.14a: 03186407.jpg.

```

File: 03186407.jpg
Standard Fields -----
Image Creation Date: 2004:07:02 17:28:49
Make: Canon
Model: Canon PowerShot SD100
Software Version: Adobe Photoshop Elements 2.0
X Resolution: 180 Pixels/Inch
Y Resolution: 180 Pixels/Inch
YCbCr Positioning: Center of Array
EXIF Fields -----
Image Capture Date: 2004:06:22 21:10:50
Image Digitized Date: 2004:06:22 21:10:50

```

Figure 2.14b: Figure 14a's image information.



Figure 2.15a: 03188831.jpg.

```
File: 03188831.jpg
Standard Fields -----
Image Creation Date: 2004:07:02 17:57:42
Make: Canon
Model: Canon PowerShot SD100
Software Version: Adobe Photoshop Elements 2.0
X Resolution: 180 Pixels/Inch
Y Resolution: 180 Pixels/Inch
YCbCr Positioning: Center of Array
EXIF Fields -----
Image Capture Date: 2004:06:19 19:49:58
Image Digitized Date: 2004:06:19 19:49:58
```

Figure 2.15b: Figure 15a's image information.



Figure 2.16a: 03241879.jpg.

```
File: 03241879.jpg
Standard Fields -----
Image Creation Date: 2004:07:02 17:40:50
Make: Canon
Model: Canon PowerShot SD100
Software Version: Adobe Photoshop Elements 2.0
X Resolution: 180 Pixels/Inch
Y Resolution: 180 Pixels/Inch
YCbCr Positioning: Center of Array
EXIF Fields -----
Image Capture Date: 2004:06:19 19:13:42
Image Digitized Date: 2004:06:19 19:13:42
```

Figure 2.16b: Figure 16a's image information.



Figure 2.17a: 03538975.jpg.

File: 03538975.jpg
 Standard Fields -----
 Image Creation Date: 2004:07:02 17:47:07
 Make: Canon
 Model: Canon PowerShot SD100
 Software Version: Adobe Photoshop Elements 2.0
 X Resolution: 180 Pixels/Inch
 Y Resolution: 180 Pixels/Inch
 YCbCr Positioning: Center of Array
 EXIF Fields -----
 Image Capture Date: 2004:06:18 23:49:04
 Image Digitized Date: 2004:06:18 23:49:04

Figure 2.17b: Figure 17a's image information.



Figure 2.18a: 05069311.jpg.

File: 05069311.jpg
 Standard Fields -----
 Image Creation Date: 2004:07:02 17:29:18
 Make: Canon
 Model: Canon PowerShot SD100
 Software Version: Adobe Photoshop Elements 2.0
 X Resolution: 180 Pixels/Inch
 Y Resolution: 180 Pixels/Inch
 YCbCr Positioning: Center of Array
 EXIF Fields -----
 Image Capture Date: 2004:06:20 18:01:44
 Image Digitized Date: 2004:06:20 18:01:44

Figure 2.18b: Figure 18a's image information.



Figure 2.19a: 05475951.jpg.

File: 05475951.jpg
 Standard Fields -----
 Image Creation Date: 2004:07:02 17:28:27
 Make: Canon
 Model: Canon PowerShot SD100
 Software Version: Adobe Photoshop Elements 2.0
 X Resolution: 180 Pixels/Inch
 Y Resolution: 180 Pixels/Inch
 YCbCr Positioning: Center of Array
 EXIF Fields -----
 Image Capture Date: 2004:06:13 21:34:00
 Image Digitized Date: 2004:06:13 21:34:00

Figure 2.19b: Figure 19a's image information



Figure 2.20a: 06139198.jpg Taken on a Canon EOS - 1DS.

```

File: 06139198.jpg
Standard Fields -----
Image Creation Date: 2003:01:29 16:14:10
Make: Canon
Model: Canon EOS-1DS
X Resolution: 72 Pixels/Inch
Y Resolution: 72 Pixels/Inch
YCbCr Positioning: Datum Point
EXIF Fields -----
Image Capture Date: 2003:01:29 16:14:10
Image Digitized Date: 2003:01:29 16:14:10
Exposure Bias: 0 EV
Focal Length: 400mm
Exposure Time: 1/250 Sec.
Aperture: f6.7
Exif Image Width: 4064 pixels
Exif Image Height: 2704 pixels
Exposure Program: Manual Control
Exposure Mode: Manual Exposure
White Balance: Auto White Balance
Metering Mode: Multi-Segment/Pattern
EXIF Version: 0220
FlashPix Version: 0100
Flash: Flash Fired;
Aperture Value: f6.7
Focal Plane X Resolution: 31750/11 Pixels/Inch
Focal Plane Y Resolution: 2704000/937 Pixels/Inch
Shutter Speed Value: 1/256 Sec.
ColorSpace: sRGB
Component Configuration: YCbCr
Scene Capture Type: Standard
Manufacturer Fields -----
Macro Mode: No Macro
Flash Mode: External
Drive Mode: Normal
Focus Mode: One-Shot
Image Size: Large
Contrast: Low
Saturation: Low
Sharpness: Low

```

Figure 2.20b: Figure 20a's image information.

```

Aperture: f6.7
Exif Image Width: 4064 pixels
Exif Image Height: 2704 pixels
Exposure Program: Manual Control
Exposure Mode: Manual Exposure
White Balance: Auto White Balance
Metering Mode: Multi-Segment/Pattern
EXIF Version: 0220
FlashPix Version: 0100
Flash: Flash Fired;
Aperture Value: f6.7
Focal Plane X Resolution: 31750/11 Pixels/Inch
Focal Plane Y Resolution: 2704000/937 Pixels/Inch
Shutter Speed Value: 1/256 Sec.
ColorSpace: sRGB
Component Configuration: YCbCr
Scene Capture Type: Standard
Manufacturer Fields -----
Macro Mode: No Macro
Flash Mode: External
Drive Mode: Normal
Focus Mode: One-Shot
Image Size: Large
Contrast: Low
Saturation: Low
Sharpness: Low
Metering Mode: Evaluative
Exposure Mode: Manual
Focal Length: 100-400mm
Flash Details: External E-TTL
White Balance: Auto
Flash Exposure Bias: 0 EV
Image Type: Canon EOS-1DS
Firmware Version: Firmware Version 1.0.1
Owner Name:
Camera Serial Number: 000136574

```

Figure 2.20c: Figure 20a's image information (further).

All images from jpg file relating to birds:



00307751.jpg



00348583.jpg



01113823.jpg



01113831.jpg



01101217.jpg



01101227.jpg



01101339.jpg



01113847.jpg



01171815.jpg



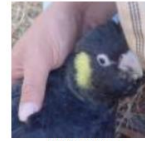
01243351.jpg



01273967.jpg



01273983.jpg



01273991.jpg



01273999.jpg



01274023.jpg



01369871.jpg



01511521.jpg



01511532.jpg



01543417.jpg



01543478.ina



01636577.ina



01636587.ina



02769337.ina



02769346.ina



02280945.jpg



02280955.jpg



02281073.jpg



02281081.jpg



02340815.jpg



02342351.jpg



02437871.jpg



02790611.jpg



02790635.jpg



02790647.jpg



02790656.jpg



02790683.jpg



02790743.jpg



02791391.jpg



02791503.jpg



02792151.ina



02792407.ina



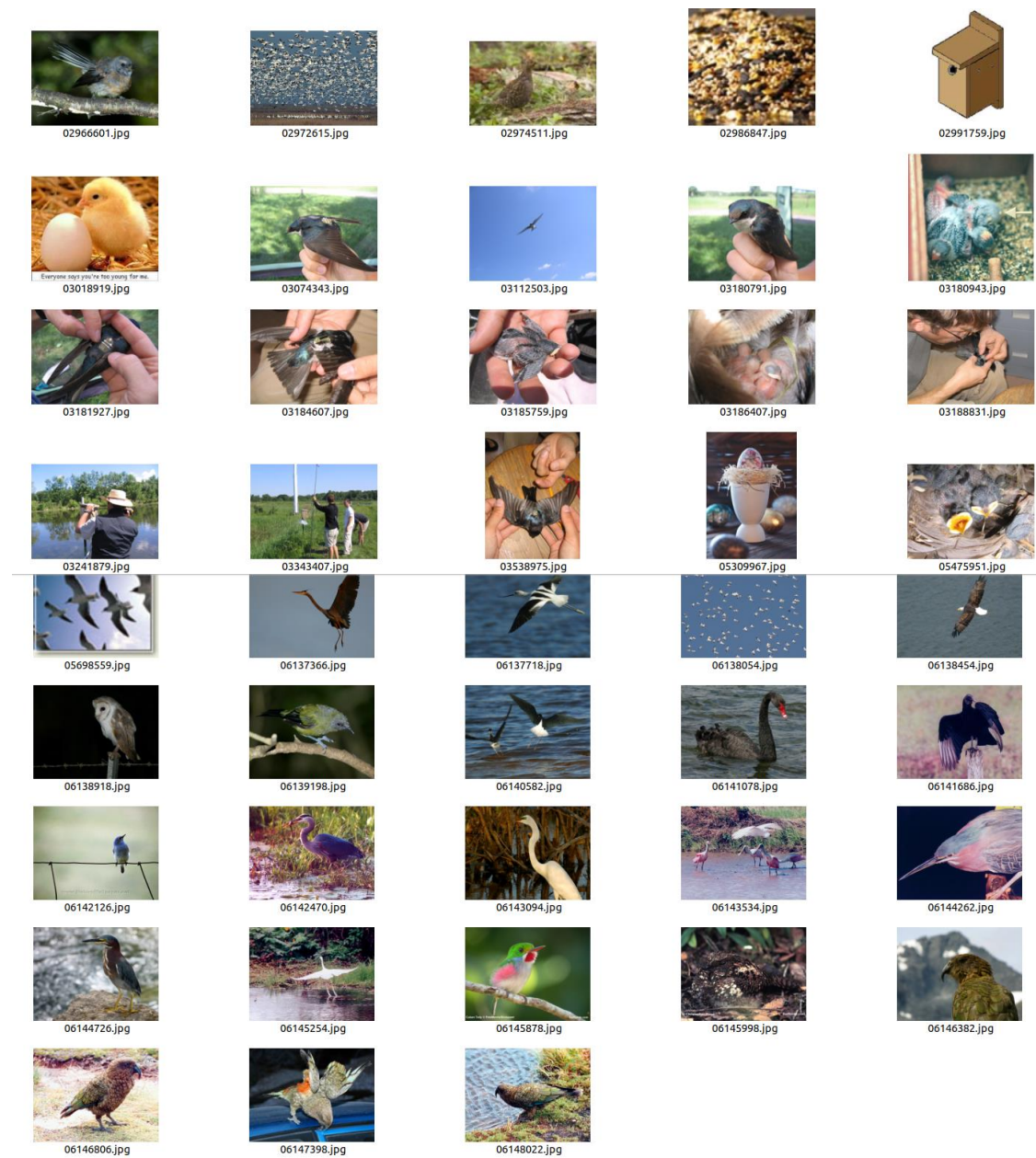
02792663.ina



02962629.ina



02962829.ina



Images Recovered from a zip file that was GPG Encrypted

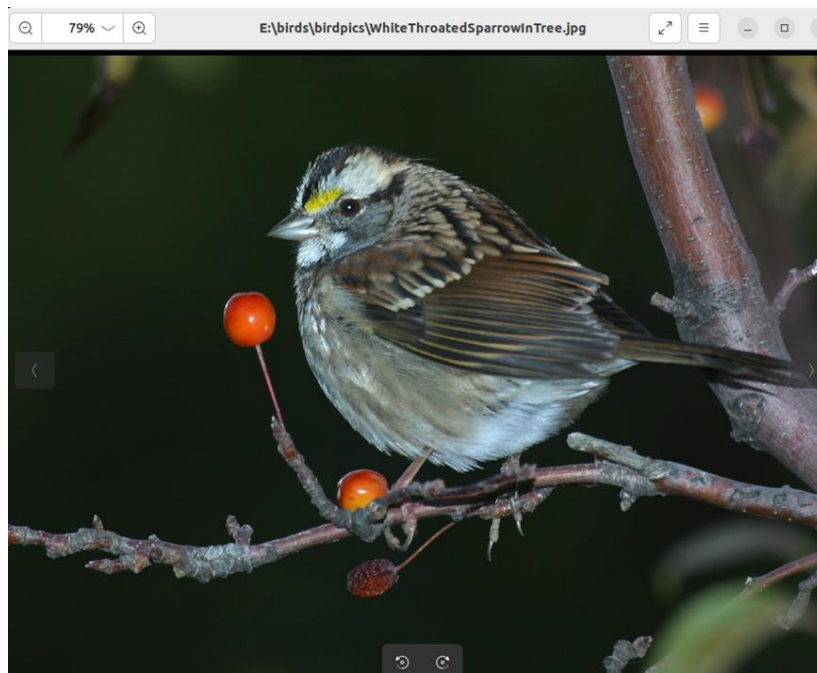


Figure 2.21: WhiteThroatedSparrowInTree.jpg

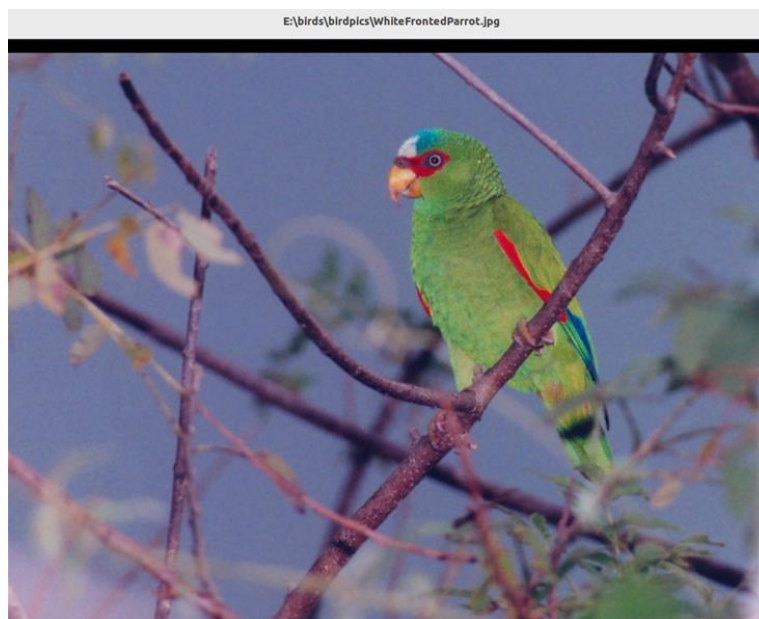


Figure 2.22: WhoteFrontedParrot.jpg

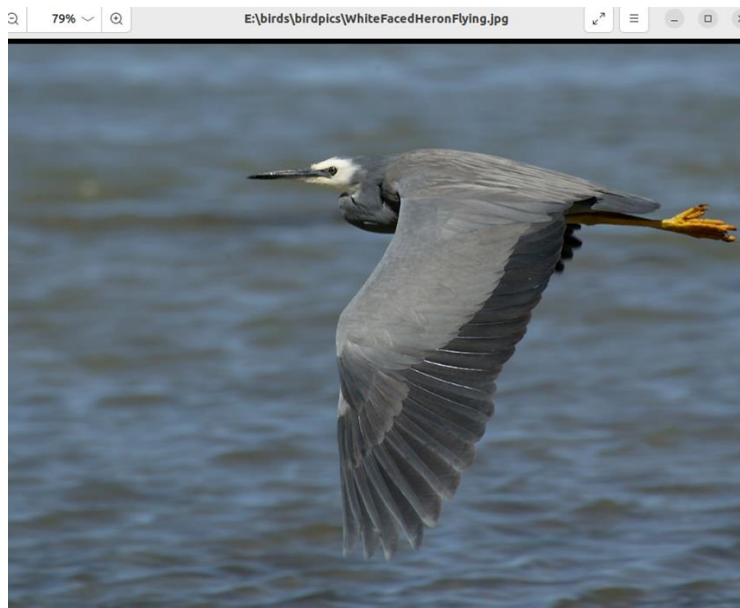


Figure 2.23: WhiteFacedHeronFlying.jpg

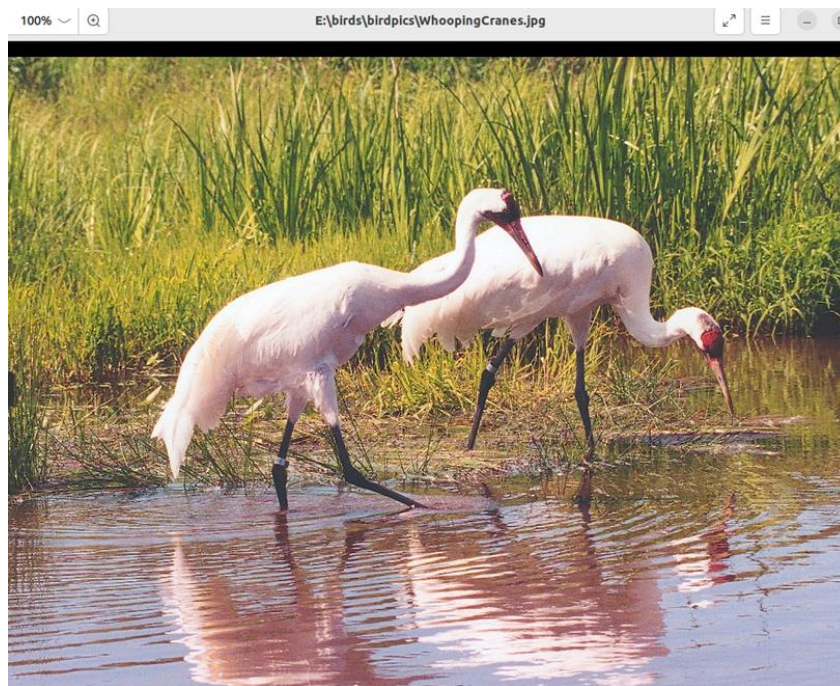


Figure 2.24: WhoopingCranes.jpg

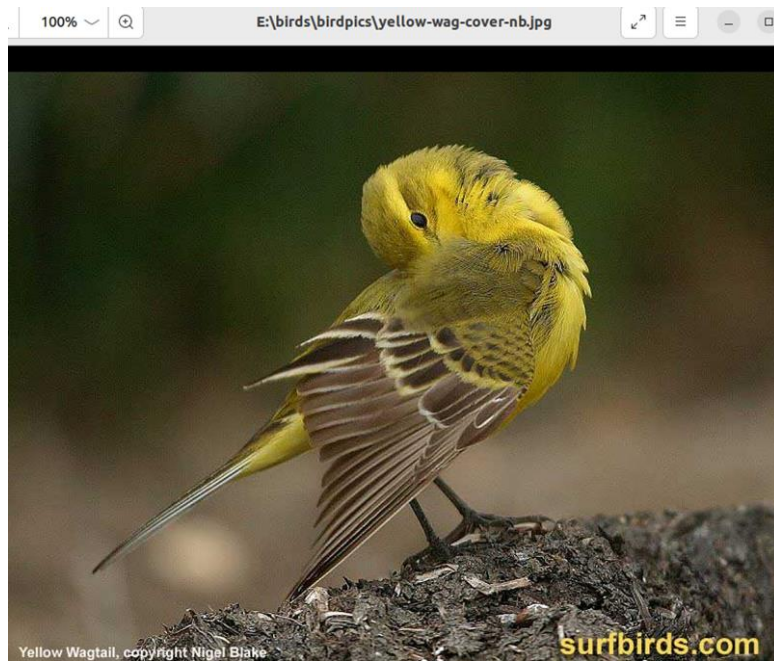


Figure 2.25: yellow-wag-cover-nb.jpg



Figure 2.26: IMG_3937_filtered.jpg



Figure 2.27: 7EYBTFLF1KAN.jpg



Figure 2.28: cute_penguin.jpg



Figure 2.28: colorful-birds.jpg



Figure 2.29: IMG_3937_filtered.jpg



Figure 2.30: glfs-storm-birds.jpg



Figure 2.31: gawall8.jpg



Figure 2.32a: BC7 feeding the birds.jpg

Item: BC7 feeding the birds.jpg
Aggregate Score: Not Notable

Analysis Result 1

Score: Not Notable
Type: EXIF Metadata
Configuration:
Conclusion:
Date Created: 2002-12-30 12:28:49 GMT
Device Make: SONY
Device Model: CYBERSHOT

Figure 2.32b: Figure 2.32a's image information



Figure 2.33: B6E058BDd01.gif



Figure 2.34: B6E0589Dd01.gif



Figure 2.35: B2906B79d01.gif



Figure 2.36: B2906B79d01.gif



Figure 2.37: B2906B79d01.gif

Images as Found Through Registry



Figure 2.46: “ready2fledge.jpg”



Figure 2.47: “newbies2.jpg



Figure 2.48: "chicks2.jpg"



Figure 2.49: "birdtrans2.jpg"



Figure 2.50: "177.jpg"



Figure 2.51: “babyscot_2weeks1.jpg”



Figure 2.52: “babyscot_vyoung.jpg”



Figure 2.53: “40m.jpg”



Figure 2.54: “wbpremium_s.jpg”



Figure 2.55: “7107298.jpg”



Figure 2.56: Full image recovered from “Doc1.doc”



Figure 2.57: Firefox Wallpaper.bmp

Appendix B – Documents

Directory Tree

- Templates (14)
- UserData (7)
- LocalService (8)
- NetworkService (8)
- Program Files (81)
- RECYCLER (3)
- System Volume Information (5)
- WINDOWS (156)
- vol9 (unallocated: 6136830-11255327)

File Views

- File Types
- Deleted Files
- MB File Size
- Data Artifacts
- Communication Accounts (5)
- E-Mail Messages (5)
- Default (Default)
- Default (5)
- Installed Programs (41)
- Metadata (33)
- Operating System Information (1)
- Recent Documents (57)
- Run Programs (87)
- Shell Bags (87)
- USB Device Attached (2)
- Web Bookmarks (12)
- Web Cookies (3)
- Web Downloads (3)
- Web History (137)

Analysis Results

- Encryption Detected (4)
- Encryption Suspected (2)
- EXIF Metadata (104)
- Extension Mismatch Detected (7)
- Keyword Hits (3813)
- User Content Suspected (104)
- OS Accounts
- Tags
- Scans

Source Name	S	C	O	E-Mail From	E-Mail To	Subject	Date Received	Message (Printed)	Message ID	Has Attachment
Inbox				ben@example.org	jdoe@example.com	good pics	2005-02-09 11:08:01 GMT	Hi thought you'd like these	Not available	Yes
Inbox				mailinglist@birds.example.com	jdoe@example.com	How to identify birds	2005-02-09 11:08:01 GMT	How to identify birds: Are you amazed at how quickly birds...	Not available	Yes
Inbox				ben@example.org	jdoe@example.com	some more good ones	2005-02-09 11:08:01 GMT	Thanks for the pics you sent me here are some I really like	Not available	Yes
Inbox				ben@example.org	jdoe@example.com	expensive birds	2005-02-09 11:08:01 GMT	A young woman was walking past a pet shop and saw an...	Not available	Yes
0000296.mbox				ben@example.org	jdoe@example.com	good pics	2005-02-09 11:08:01 GMT	Hi thought you'd like these	Not available	Yes

Data Content

Hex Text Application Source File Metadata OS Account Data Artifacts Analysis Results Context Annotations Other Occurrences

Result: 3 of 7 Result

E-Mail Messages

From: ben@example.org
To: jdoe@example.com
CC:
Subject: good pics

Headers Text HTML PDF Attachments (3) Accounts

Hi thought you'd like these

enjoy

Figure 3.1: Autopsy email received by jdoe@example.com

```

1 # HTTP Cookie File
2 # http://www.netscape.com/newsref/std/cookie_spec.html
3 # This is a generated file! Do not edit.
4 # To delete cookies, use the Cookie Manager.
5
6 .google.co.uk TRUE / FALSE 2147368447 PREF ID=61669b956c7e0f5a:LD=en:TM=1106584100:LH=1106584100:S=hh0ICG0xscdQrph
7 .bbc.co.uk TRUE / FALSE 1233499441 BBC-UID
8 .bbc.co.uk TRUE / FALSE 1138891537 B8CMediaSelector nm3Armk2obk3Abb
9 .google.com TRUE / FALSE 2147368447 PREF ID=ad3ac92f0f40d59a:CH=1:TM=1106584100:LH=1106584100:S=H0nW0MkUnY51BF
10 .focalex.com TRUE / FALSE 1422802881 focalEx_aa_p os&ld8aa09m8CEURPk5MKW65C2A
11 .fastclick.net TRUE / FALSE 1169561752 pluto 663483068|0|66b5a384-3d2e-44e2-9a8b-44094f4503db|
12 .fastclick.net TRUE / FALSE 1108047069 n2 18809:1:45368:8:1107442077|
13 www.adobe.com FALSE / FALSE 1422723126 AMID 130:159:248:44:175491107363127998
14 .112.207.net TRUE / FALSE 1265037880 s_vl_tthenx7cxhdx7Ex60 [CS]v4|4200E73600006854-A000BA400000001|4200E735[CE]
15 .real.com TRUE / FALSE 1112539547 RNsites forns03-00k8Y:33
16 .indb.com TRUE / FALSE 2145743999 uu 0N3KNigtbcjldc0BGF8CwGVHxv2w:1NhZNWw/QQr:5JjQVn3bpc-KRBVn/tWY7odcPEKLUGf53h9Kpd4rFN2zP33dvSNp3b8gF474WQ7q+fgG+YtZB/
17 .advertising.com TRUE / FALSE 1265035301 ROLL cLZBHGZb1BT4K3vhdH5XYloZoz+TN3K1
18 .advertising.com TRUE / FALSE 1265035226 ACID bb0900110735522600101
19 .servedby.advertising.com TRUE / FALSE 1109947301 65413475 _4200e6a5,6431706035,684689*223368_
20 .servedby.advertising.com TRUE / FALSE 1109947226 19410806 _4200e65a,3229376194,689639*244838_
21 .servedby.advertising.com TRUE / FALSE 1109947230 80141706 _4200e65e,2726379568,689642*233315_
22 .servedby.advertising.com TRUE / FALSE 1109947268 82088981 _4200e684,9079152656,684689*242879_
23 .advertising.com TRUE / FALSE 1265035401 BASE ns1g0BNakPyPLDr+EQ5Kr/cJb/wrU5Fe0qZ51Nl
24 .servedby.advertising.com TRUE / FALSE 1109947404 15326560 _4200e70c,5539389421,689991*232484_
25 .servedby.advertising.com TRUE / FALSE 1109947401 72136016 _4200e709,9535359602,689991*242222_
26 .servedby.advertising.com TRUE / FALSE 1109947403 23536105 _4200e70b,2209937515,689989*244838_
27 .maxservicing.com TRUE / FALSE 1422715249 uu en923GpWRYtT
28 .videodetective.com FALSE / FALSE 1138891329 videokbrate 300
29 www.hatth.com FALSE / FALSE 1609372654 session
30 .firstx5FvIsited5SFrerefererKSFld=BIROINGK2EABOUTN2ECOMHrefererKSFld=BIROINGK2EABOUTN2ECOMHdateK5Fcreated=2K2F2K2F2005+2K3A29K3A33+PMAId=XXU6S831EKSaHTN23MHTSDQN39QK61V
31 www.hatth.com FALSE / FALSE 1108166254 sessionK5Fsrc src=
32 .about.com TRUE / FALSE 1107958967 Amc N
33 .about.com TRUE / FALSE 1107959203 gs birdng
34 .about.com TRUE / FALSE 7107440743 zFO 52280520810500801
35 .about.com TRUE / FALSE 1138890403 referrer http://about.com
36 .about.com TRUE / FALSE 1138890403 trallblazer referrer-http://birding.about.com/library/weekly/aa010703a.htm
37 .about.com TRUE / FALSE 1169562189 TID 0mfhts21101ole
38 .about.com TRUE / FALSE 1707354099 jsc 13
39 .about.com TRUE / FALSE 2145916799 Thog 522EMF00205A1B0c
40 .birding.about.com TRUE / FALSE 7107440743 zFS 52280520810500801
41 .audiencecatch.net TRUE / FALSE 1138458167 TID 0mfhts21101pog
42 .doubleclick.net TRUE / FALSE 1201962166 ld 8000004c8c97582
43 .atdnt.com TRUE / FALSE 1264982398 AA002 001107354287-989877906/1108563887
44 .shop.com TRUE / FALSE 2145916801 AMID: 319429603

```

Figure 3.2: Firefox “cookies.txt” (1/2)

```

44 .casalemedia.com TRUE / FALSE 1109927756 CHX2 40842811073533757K53524N43608
45 .casalemedia.com TRUE / FALSE 1138094156 CHD 41W9KUS1DMACZ0ItgAAAAAN
46 www.relnaxtop.com FALSE / FALSE 1422713927 vlsitor_id 96988879
47 .amazon.co.uk TRUE / FALSE 2082758401 ubid-acbuk 432-0491950-8565500
48 www.googleadservices.com FALSE / FALSE /pagead/conversion/1072099989/ FALSE 1109945507 Conversion
49 .office.microsoft.com TRUE / FALSE 1138274120 awssuserguid guid=45a3adcc-4fa0-4e1b-b6a4-ecd3400df84c

```

Figure 3.3: Firefox “cookies.txt” (2/2)

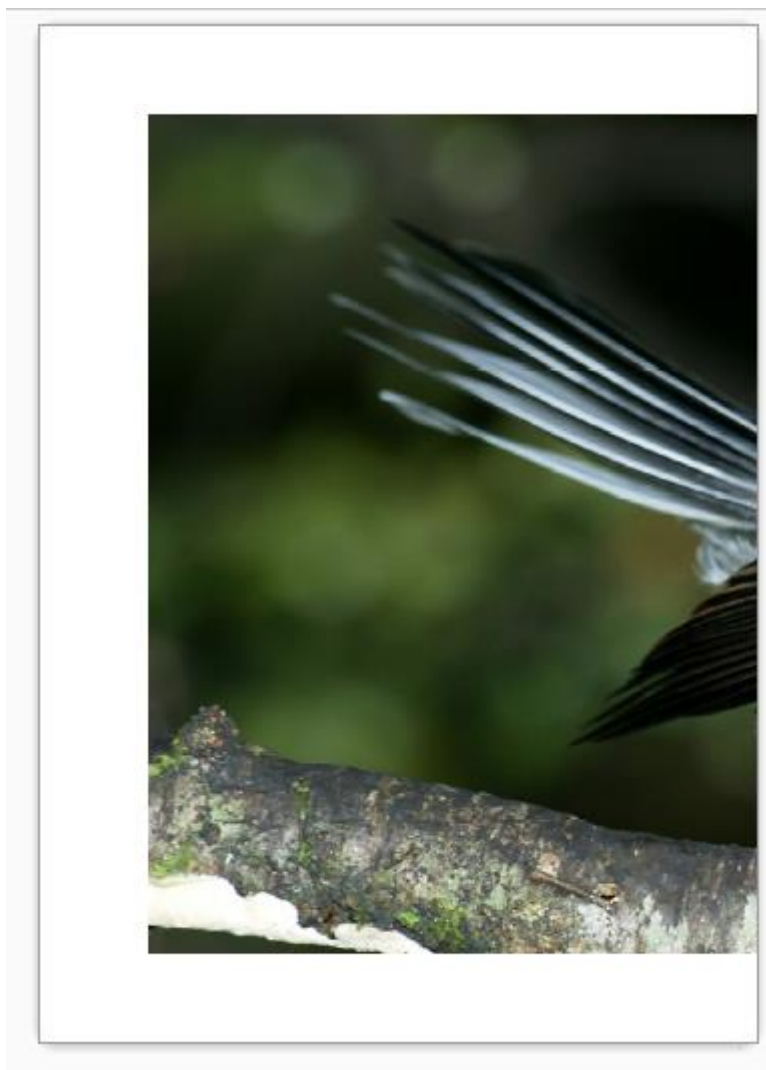


Figure 3.4: “Doc1.doc”

MZHistoryView - C:\Users\Owner\Desktop\JD SHARED EVIDENCE\Mozilla\Firefox\Profiles\w4nf3jbb1.default\history.dat										
File Edit View Options Help										
URL	First Visit Date	Last Visit Date	Visit Count	Referrer	Host Name	Title	Record Ind...	Visit Type	Frecency	URL Length
http://start.mozilla.org/firefox?client=firefox-a&ut...	24/01/2005 16:28:20	02/02/2005 14:10:48	2		start.mozilla.org		0	0	80	
http://www.google.com/firefox?client=firefox-a&ut...	24/01/2005 16:28:20	02/02/2005 14:10:48	2		google.com		1	0	77	
http://www.google.co.uk/crfer?c=PREP&3D:TM&...	24/01/2005 16:28:20	24/01/2005 16:28:20	0		google.co.uk		2	0	146	
http://www.google.co.uk/firefox?client=firefox-a&ut...	24/01/2005 16:28:20	02/02/2005 14:10:48	2		google.co.uk	Mozilla Firefox Start Page	3	0	79	
http://www.mozilla.org/about/	24/01/2005 16:28:37	24/01/2005 16:28:37	0		mozilla.org	About Mozilla	4	0	29	
http://www.mozilla.org/products/	24/01/2005 16:28:40	24/01/2005 16:28:40	0		mozilla.org	Mozilla Products	5	0	32	
http://www.mozilla.org/products/thunderbird/	24/01/2005 16:28:40	24/01/2005 16:28:40	0		mozilla.org	Thunderbird - Reclaim Your In...	6	0	44	
http://download.mozilla.org/?product=thunderbir...	24/01/2005 16:28:47	24/01/2005 16:28:47	0		download.mozilla...		7	0	66	
ftp://ftp.scarlet.be/pub/mozilla.org/thunderbird/r...	24/01/2005 16:28:47	24/01/2005 16:28:47	0		ftp.scarlet.be		8	0	103	
http://r.office.microsoft.com/r/tidOfficeUpdate?c...	25/01/2005 11:15:19	25/01/2005 11:15:19	0		r.office.microsoft...		9	0	58	
http://office.microsoft.com/officeupdate	25/01/2005 11:15:19	25/01/2005 11:15:19	0		office.microsoft.c...		10	0	40	
http://office.microsoft.com/officeupdate/	25/01/2005 11:15:19	25/01/2005 11:15:19	0		office.microsoft.c...		11	0	41	
http://office.microsoft.com/en-gb/officeupdate/d...	25/01/2005 11:15:19	25/01/2005 11:15:19	0		office.microsoft.c...	Microsoft Office Downloads ...	12	0	59	
http://office.microsoft.com/search/redir.aspx?As...	25/01/2005 11:15:40	25/01/2005 11:15:40	0		office.microsoft.c...		13	0	96	
http://office.microsoft.com/officeupdate/maincat...	25/01/2005 11:15:40	25/01/2005 11:15:40	0		office.microsoft.c...		14	0	66	
http://office.microsoft.com/en-gb/FX01053255103...	25/01/2005 11:15:40	25/01/2005 11:15:40	0		office.microsoft.c...	Microsoft Office Online: Inter...	15	0	53	
http://office.microsoft.com/search/redir.aspx?As...	25/01/2005 11:15:46	25/01/2005 11:15:46	0		office.microsoft.c...		16	0	96	
http://office.microsoft.com/en-gb/officeupdate/...	25/01/2005 11:15:46	25/01/2005 11:15:46	0		office.microsoft.c...	Microsoft Office Downloads ...	17	0	66	
http://www.google.co.uk/search?client=firefox-a&...	02/02/2005 14:11:46	02/02/2005 14:11:46	0		google.co.uk	Google Search: birds	18	0	123	
http://www.pbs.org/lifeofbirds/	02/02/2005 14:11:54	02/02/2005 14:11:54	2		pbs.org	The Life of Birds	19	0	31	
http://www.pbs.org/lifeofbirds/songs/index.html	02/02/2005 14:12:05	02/02/2005 15:11:34	2		pbs.org	The Life of Birds Songs	20	0	47	
http://www.google.co.uk/pagead/clk?adufl=...	02/02/2005 14:14:49	02/02/2005 14:14:49	0		google.co.uk		21	0	291	
http://www.googleadservices.com/pagead/adclk...	02/02/2005 14:14:49	02/02/2005 14:14:49	0		googleadservices...		22	0	302	
http://www.amazon.co.uk/ece/obidos/external-s...	02/02/2005 14:14:49	02/02/2005 14:14:49	0		amazon.co.uk		23	0	98	
http://www.amazon.co.uk/ece/obidos/external-s...	02/02/2005 14:14:49	02/02/2005 14:14:49	0		amazon.co.uk	Amazon.co.uk: Search Results...	24	0	118	
http://www.amazon.co.uk/ece/obidos/ASIN/000...	02/02/2005 14:14:59	02/02/2005 14:14:59	0		amazon.co.uk	Amazon.co.uk: Books: Garden...	25	0	107	
http://www.amazon.co.uk/ece/obidos/ASIN/071...	02/02/2005 14:15:29	02/02/2005 14:15:29	0		amazon.co.uk	Amazon.co.uk: Books: The Se...	26	0	107	
http://www.google.co.uk/search?c=bird+wallpap...	02/02/2005 14:15:42	02/02/2005 14:15:42	0		google.co.uk	Google Search: bird wallpaper	27	0	153	
http://www.naturewallpaper.net/birds_L.html	02/02/2005 14:15:51	02/02/2005 14:15:51	0		naturewallpaper.net	Free Bird Wallpaper - Bald Eag...	28	0	43	
http://as.casalemedia.com/s?5=535248u=http%3...	02/02/2005 14:15:51	02/02/2005 14:15:51	0		as.casalemedia.com		29	0	108	
http://pagead2.googlesyndication.com/pagead/a...	02/02/2005 14:15:56	02/02/2005 14:15:56	0		pagead2.googlesy...		30	0	472	
http://asg36.casalemedia.com/s?5=535248u=http...	02/02/2005 14:15:56	02/02/2005 14:15:56	0		asg36.casalemedia...		31	0	111	
http://sig10.casalemedia.com/V2/40842/43608	02/02/2005 14:15:56	02/02/2005 14:15:56	0		sig10.casalemedia...		32	0	43	
http://sig10.casalemedia.com/V2/40842/43608/	02/02/2005 14:15:56	02/02/2005 14:15:56	0		sig10.casalemedia...	3-Home	33	0	44	
http://www.naturewallpaper.net/birds4pages/ima...	02/02/2005 14:16:21	02/02/2005 14:16:21	0		naturewallpaper.net	bird_eagle3.jpg	34	0	54	
http://media.fastclick.net/vi/getmedia?n=8&id=...	02/02/2005 14:16:21	02/02/2005 14:16:21	0		media.fastclick.net	Winner !!!	35	0	125	
http://pagead2.googlesyndication.com/pagead/a...	02/02/2005 14:16:21	02/02/2005 14:16:21	0		pagead2.googlesy...		36	0	337	
http://www.naturewallpaper.net/birds4pages/ima...	02/02/2005 14:18:45	02/02/2005 14:18:45	0		naturewallpaper.net	snow_geese.jpg	37	0	55	
http://media.fastclick.net/vi/getmedia?n=8&id=...	02/02/2005 14:18:45	02/02/2005 14:18:45	0		media.fastclick.net	Winner !!!	38	0	125	
http://pagead2.googlesyndication.com/pagead/a...	02/02/2005 14:18:45	02/02/2005 14:18:45	0		pagead2.googlesy...		39	0	338	
http://pagead2.googlesyndication.com/pagead/r/c...	02/02/2005 14:19:48	02/02/2005 14:19:48	0		pagead2.googlesy...		40	0	384	
http://pagead2.googlesyndication.com/pagead/a...	02/02/2005 14:19:48	02/02/2005 14:19:48	0		pagead2.googlesy...		41	0	387	
http://uk.shop.com/amos/cc/main/cc/catalog/...	02/02/2005 14:19:56	02/02/2005 14:19:56	0		uk.shop.com	SHOP.COM - Search results fo...	42	0	96	
http://amos.shop.com/amos/cc/main/cc/catalog/...	02/02/2005 14:20:19	02/02/2005 14:20:19	0		amos.shop.com	SHOP.COM - Bird Safari - Bird...	43	0	149	
http://www.google.co.uk/search?hl=en&client=fin...	02/02/2005 14:22:25	02/02/2005 14:22:25	0		google.co.uk	Google Search: bird stories	44	0	121	
http://birding.about.com/od/storie/	02/02/2005 14:22:38	02/02/2005 14:22:38	0		birding.about.com	Bird Stories and Tales	45	0	36	
http://n479ad.doubleclick.net/abt/hobbies/no...	02/02/2005 14:22:38	02/02/2005 14:22:38	0		n479ad.doublecl...	Click Here!	46	0	195	
http://z.about.com/5/ad/go.html?gs=birding	02/02/2005 14:22:38	02/02/2005 14:26:43	8		z.about.com	Google Offers	47	0	41	

Figure 3.5: Firefox “history.dat” (1/4)

http://z.about.com/0/ip/417/5.htm?CDN=hobbies	02/02/2005 14:22:38	02/02/2005 14:26:43	8		z.about.com		48	0	45	
http://z.about.com/0/ip/496/0.htm	02/02/2005 14:22:38	02/02/2005 14:26:43	8		z.about.com	Untitled Document	49	0	33	
http://z.about.com/0/ip/512/6.htm	02/02/2005 14:22:38	02/02/2005 14:26:43	8		z.about.com		50	0	33	
http://z.about.com/0/ip/512/12.htm	02/02/2005 14:22:38	02/02/2005 14:26:43	8		z.about.com		51	0	34	
http://n479ad.doubleclick.net/abt/hobbies/no...	02/02/2005 14:22:45	02/02/2005 14:22:45	0		n479ad.doublecl...	Click Here!	52	0	185	
http://z.about.com/6/o/b/html?gs=birding	02/02/2005 14:22:45	02/02/2005 14:26:43	7		z.about.com	Current Headlines	53	0	39	
http://z.about.com/5/ad/am.htm?pid=about	02/02/2005 14:22:45	02/02/2005 14:26:43	8		z.about.com	Audience Match Data Agent	54	0	40	
http://birding.about.com/library/weekly/aa102598...	02/02/2005 14:23:04	02/02/2005 14:23:04	0		birding.about.com	Wings of Terror - avian humor...	55	0	52	
http://n479ad.doubleclick.net/abt/hobbies/no...	02/02/2005 14:23:04	02/02/2005 14:23:04	0		n479ad.doublecl...	Click Here!	56	0	194	
http://z.about.com/7/o/cv.html?gs=birding	02/02/2005 14:23:04	02/02/2005 14:26:43	6		z.about.com		57	0	40	
http://z.about.com/5/o/c.htm?gs=birding	02/02/2005 14:23:04	02/02/2005 14:26:43	6		z.about.com	Related Articles	58	0	39	
http://n479ad.doubleclick.net/abt/hobbies/no...	02/02/2005 14:23:04	02/02/2005 14:23:04	0		n479ad.doublecl...	Click Here!	59	0	185	
http://n479ad.doubleclick.net/abt/hobbies/no...	02/02/2005 14:23:04	02/02/2005 14:23:04	0		n479ad.doublecl...	Click Here!	60	0	184	
http://clk.about.com/?zjs=18/y/8&sdns=hobbies_bi...	02/02/2005 14:23:56	02/02/2005 14:23:56	0		clk.about.com		61	0	54	
http://birding.about.com/library/bialphaypsefbi...	02/02/2005 14:23:56	02/02/2005 14:23:56	0		birding.about.com	Alphabetical Index of Birds	62	0	54	
http://n479ad.doubleclick.net/abt/hobbies/no...	02/02/2005 14:23:56	02/02/2005 14:23:56	0		n479ad.doublecl...	Click Here!	63	0	212	
http://n479ad.doubleclick.net/abt/hobbies/no...	02/02/2005 14:23:56	02/02/2005 14:23:56	0		n479ad.doublecl...	Click Here!	64	0	203	
http://n479ad.doubleclick.net/abt/hobbies/no...	02/02/2005 14:23:56	02/02/2005 14:23:56	0		n479ad.doublecl...	Click Here!	65	0	202	
http://clk.about.com/?zjs=18/y/8&sdns=hobbies_bi...	02/02/2005 14:24:18	02/02/2005 14:24:18	0		clk.about.com		66	0	154	
http://birding.about.com/library/bihousespecs.htm	02/02/2005 14:24:18	02/02/2005 14:24:18	0		birding.about.com	Specs for Building Birdhouses ...	67	0	49	
http://n479ad.doubleclick.net/abt/hobbies/no...	02/02/2005 14:24:18	02/02/2005 14:24:18	0		n479ad.doublecl...	Click Here!	68	0	212	
http://n479ad.doubleclick.net/abt/hobbies/no...	02/02/2005 14:24:18	02/02/2005 14:24:18	0		n479ad.doublecl...	Click Here!	69	0	203	
http://n479ad.doubleclick.net/abt/hobbies/no...	02/02/2005 14:24:18	02/02/2005 14:24:18	0		n479ad.doublecl...	Click Here!	70	0	202	
http://birding.about.com/cs/buildhouses/index.ht...	02/02/2005 14:24:32	02/02/2005 14:24:32	0		birding.about.com		71	0	49	
http://birding.about.com/cs/buildhouses/	02/02/2005 14:24:32	02/02/2005 14:24:32	0		birding.about.com	How to Build Bird Houses and...	72	0	40	
http://n479ad.doubleclick.net/abt/hobbies/no...	02/02/2005 14:24:32	02/02/2005 14:24:32	0		n479ad.doublecl...	Click Here!	73	0	213	
http://n479ad.doubleclick.net/abt/hobbies/no...	02/02/2005 14:24:32	02/02/2005 14:24:32	0		n479ad.doublecl...	Click Here	74	0	204	
http://n479ad.doubleclick.net/abt/hobbies/no...	02/02/2005 14:24:32	02/02/2005 14:24:32	0		n479ad.doublecl...	Click Here!	75	0	203	
http://birding.about.com/cs/birdhousefeeders/a/...	02/02/2005 14:24:42	02/02/2005 14:24:42	0		birding.about.com	Free Feeder and Birdhouse Pla...	76	0	65	
http://n479ad.doubleclick.net/abt/hobbies/no...	02/02/2005 14:24:42	02/02/2005 14:24:42	0		n479ad.doublecl...	Click Here!	77	0	212	
http://n479ad.doubleclick.net/abt/hobbies/no...	02/02/2005 14:24:42	02/02/2005 14:24:42	0		n479ad.doublecl...	Click Here!	78	0	201	
http://n479ad.doubleclick.net/abt/hobbies/no...	02/02/2005 14:24:42	02/02/2005 14:24:42	0		n479ad.doublecl...	Click Here!	79	0	202	
http://view.atdmt.com/VON/view/btcmvov0200...	02/02/2005 14:24:42	02/02/2005 14:24:42	0		view.atdmt.com	vorange_vpart_gethruibiz_336	80	0	251	
http://birding.about.com/library/weekly/aa10703...	02/02/2005 14:24:55	02/02/2005 14:24:55	0		birding.about.com	Birding and Birdwatching - Bu...	81	0	53	
http://n479ad.doubleclick.net/abt/hobbies/no...	02/02/2005 14:24:55	02/02/2005 14:24:55	0		n479ad.doublecl...	Click Here!	82	0	212	
http://n479ad.doubleclick.net/abt/hobbies/no...	02/02/2005 14:24:55	02/02/2005 14:24:55	0		n479ad.doublecl...	Click Here!	83	0	203	
http://n479ad.doubleclick.net/abt/hobbies/no...	02/02/2005 14:24:55	02/02/2005 14:24:55	0		n479ad.doublecl...	Click Here!	84	0	202	
http://clk.about.com/?zjs=18/y/8&sdns=hobbies_bi...	02/02/2005 14:26:43	02/02/2005 14:26:43	0		clk.about.com		85	0	161	
http://birding.about.com/library/bihousefeeders.p...	02/02/2005 14:26:43	02/02/2005 14:26:43	0		birding.about.com	Birding and Birdwatching - Pa...	86	0	55	
http://n479ad.doubleclick.net/abt/hobbies/no...	02/02/2005 14:26:43	02/02/2005 14:26:43	0		n479ad.doublecl...	Click Here!	87	0	212	
http://n479ad.doubleclick.net/abt/hobbies/no...	02/02/2005 14:26:43	02/02/2005 14:26:43	0		n479ad.doublecl...	Click Here!	88	0	203	
http://n479ad.doubleclick.net/abt/hobbies/no...	02/02/2005 14:26:43	02/02/2005 14:26:43	0		n479ad.doublecl...	Click Here!	89	0	202	
http://pagead2.googlesyndication.com/pagead/r/c...	02/02/2005 14:27:01	02/02/2005 14:27:01	0		pagead2.googlesy...		90	0	327	
http://pagead2.googlesyndication.com/pagead/a...	02/02/2005 14:27:01	02/02/2005 14:27:01	0		pagead2.googlesy...		91	0	330	
http://www.haiths.com/	02/02/2005 14:27:07	02/02/2005 14:27:07	0		haiths.com	Haith's Wild Bird Foods & Bill ...	92	0	22	
http://www.haiths.com/Features.asp?lookup=08b...	02/02/2005 14:28:31	02/02/2005 14:28:31	0		haiths.com	Haith's Wild Bird Foods & Bill ...	93	0	118	
http://www.imdb.com/title/tt0056869/	02/02/2005 14:40:24	02/02/2005 14:40:24	0		imdb.com	The Birds 1963	94	0	36	

Figure 3.6: Firefox “history.dat” (2/4)

http://www.imdb.com/google/box?num=3;k=po...	02/02/2005 14:40:30	02/02/2005 14:40:30	0	imdb.com		95	0	137
http://www.imdb.com/rg/title-tease/trailers/titl...	02/02/2005 14:40:46	02/02/2005 14:40:46	0	imdb.com		96	0	68
http://www.imdb.com/title/t0056869/trailers	02/02/2005 14:40:46	02/02/2005 14:40:46	0	imdb.com	Trailers for The Birds 1963	97	0	44
http://rcm.amazon.com/e/cm?tf=ifr&t=imdb-rec...	02/02/2005 14:40:46	02/02/2005 14:40:46	0	rcm.amazon.com		98	0	95
http://www.imdb.com/google/box?num=3;k=po...	02/02/2005 14:40:46	02/02/2005 14:40:46	0	imdb.com		99	0	132
http://www.imdb.com/google/box?num=3;k=ma...	02/02/2005 14:40:49	02/02/2005 14:40:49	0	imdb.com		100	0	119
http://www.imdb.com/rg/TITLEPA_VIDEOS?/http...	02/02/2005 14:41:00	02/02/2005 14:41:00	0	imdb.com		101	0	91
http://videodetective.com/home.asp?PublishedI...	02/02/2005 14:41:00	02/02/2005 14:41:00	0	videodetective.com	VideoDetective.com - Movie T...	102	0	51
http://videodetective.com/player.asp?PublishedI...	02/02/2005 14:41:56	02/02/2005 14:41:56	0	videodetective.com		103	0	89
http://videodetective.com/test.asp?list=307629f83...	02/02/2005 14:41:56	02/02/2005 14:41:56	0	videodetective.com		104	0	100
http://videodetective.com/test.asp?list=307629f83...	02/02/2005 14:41:56	02/02/2005 14:41:56	0	videodetective.com		105	0	115
http://videodetective.com/choosespeed.asp?Brow...	02/02/2005 14:42:00	02/02/2005 14:42:00	0	videodetective.com	Video Detective Player	106	0	191
http://videodetective.com/player.asp?PublishedI...	02/02/2005 14:42:09	02/02/2005 14:42:09	0	videodetective.com	VideoDetective.com - Player	107	0	105
http://videodetective.com/play.asp?New=True&Li...	02/02/2005 14:42:09	02/02/2005 14:42:09	0	videodetective.com		108	0	131
http://videodetective.com/titleinfo.asp	02/02/2005 14:42:09	02/02/2005 14:42:09	0	videodetective.com		109	0	39
http://videodetective.com/Playlist.asp?	02/02/2005 14:42:09	02/02/2005 14:42:09	0	videodetective.com		110	0	39
http://news.bbc.co.uk/	02/02/2005 14:44:01	02/02/2005 14:44:01	2	news.bbc.co.uk	BBC NEWS News Front Page	111	0	22
http://news.bbc.co.uk/nol/ukfs_news/hi/front_pa...	02/02/2005 14:44:01	02/02/2005 14:44:01	0	news.bbc.co.uk		112	0	60
http://news.bbc.co.uk/1/hi/uk_politics/4227829.stm	02/02/2005 14:44:12	02/02/2005 14:44:12	0	news.bbc.co.uk	BBC NEWS Politics Blair def...	113	0	50
http://news.bbc.co.uk/1/hi/uk_politics/4227829.st...	02/02/2005 14:45:28	02/02/2005 14:45:28	0	news.bbc.co.uk		114	0	51
http://www.bbc.co.uk/mediaselector/check/nolv...	02/02/2005 14:45:28	02/02/2005 14:45:28	0	bbc.co.uk	BBC Media Selector	115	0	139
http://www.bbc.co.uk/cgi-perl/mediaselector/apg...	02/02/2005 14:45:37	02/02/2005 14:45:37	0	bbc.co.uk	BBC Media Selector	116	0	260
http://www.bbc.co.uk/mediaselector/check/nolv...	02/02/2005 14:45:37	02/02/2005 14:45:37	0	bbc.co.uk		117	0	190
http://news.bbc.co.uk/nolavconsole/ukfs_news/hi...	02/02/2005 14:45:37	02/02/2005 14:45:37	0	news.bbc.co.uk	BBC News Player	118	0	154
http://news.bbc.co.uk/nolavconsole/ukfs_news/hi...	02/02/2005 14:45:37	02/02/2005 14:45:37	0	news.bbc.co.uk		119	0	75
http://news.bbc.co.uk/nolavconsole/shared/play...	02/02/2005 14:45:37	02/02/2005 14:45:37	0	news.bbc.co.uk	Untitled	120	0	149
http://news.bbc.co.uk/nolavconsole/ukfs_news/hi...	02/02/2005 14:45:37	02/02/2005 14:45:37	0	news.bbc.co.uk	BBC NEWS UK Blair defends...	121	0	95
http://news.bbc.co.uk/nolavconsole/ukfs_news/hi...	02/02/2005 14:45:37	02/02/2005 14:45:37	0	news.bbc.co.uk		122	0	74
http://news.bbc.co.uk/nolavconsole/shared/play...	02/02/2005 14:45:37	02/02/2005 15:04:55	2	news.bbc.co.uk	Untitled	123	0	198
http://news.bbc.co.uk/nolavconsole/ukfs_news/hi...	02/02/2005 14:45:37	02/02/2005 15:04:55	2	news.bbc.co.uk		124	0	67
http://news.bbc.co.uk/nolavconsole/ukfs_news/hi...	02/02/2005 14:45:37	02/02/2005 15:04:55	2	news.bbc.co.uk	Untitled	125	0	66
http://news.bbc.co.uk/1/hi/scotland/default.stm	02/02/2005 15:11:19	02/02/2005 15:11:19	0	news.bbc.co.uk	BBC NEWS Scotland	126	0	47
http://www.pbs.org/lifeofbirds/songs/kakapo.ram	02/02/2005 15:11:41	02/02/2005 15:11:41	0	pbs.org		127	0	47
http://www.pbs.org/lifeofbirds/songs/dawn.ram	02/02/2005 15:12:06	02/02/2005 15:12:06	0	pbs.org		128	0	45
http://www.google.co.uk/search?q=windows+gn...	02/02/2005 15:57:40	02/02/2005 15:57:40	0	google.co.uk	Google Search: windows gnupg	129	0	152
http://www.gnupg.org/download.html	02/02/2005 15:57:51	02/02/2005 15:57:51	0	gnupg.org		130	0	34
http://www.gnupg.org/download/	02/02/2005 15:57:51	02/02/2005 15:57:51	0	gnupg.org	Download - GnuPG.org	131	0	30
http://www.winpt.org/	02/02/2005 15:58:23	02/02/2005 15:58:23	0	winpt.org	WinPT	132	0	21
http://www.stud.uni-hannover.de/~twoaday/win...	02/02/2005 15:58:23	02/02/2005 15:58:23	0	stud.uni-hannover...		133	0	51
http://www.google.co.uk/search?q=windows+gn...	02/02/2005 15:58:48	02/02/2005 15:58:48	0	google.co.uk	Google Search: windows gnupg	134	0	118
http://openpgp.vie-privet.org/wingpg.html	02/02/2005 15:58:58	02/02/2005 15:58:58	0	openpgp.vie-privet...	GnuPG Windows	135	0	41
http://shell.windows.com/Filesassoc/0409/xml/redi...	02/02/2005 16:51:54	02/02/2005 16:51:54	0	shell.windows.com	Microsoft Application Search	136	0	61
http://shell.windows.com/Filesassoc/0409/Search...	02/02/2005 16:51:54	02/02/2005 16:51:54	0	shell.windows.com		137	0	56
http://www.adobe.com/products/acrobat/readste...	02/02/2005 16:52:06	02/02/2005 16:52:06	0	adobe.com	Adobe Reader - Download	138	0	52
http://www.adobe.com/products/acrobat/readste...	02/02/2005 16:52:50	02/02/2005 16:52:50	0	adobe.com	Adobe Reader - Download Th...	139	0	194
http://ardownload.adobe.com/pub/adobe/reader...	02/02/2005 16:53:13	02/02/2005 16:53:13	0	ardownload.adobe...		140	0	82
http://www.google.co.uk/search?client=firefox-a...	03/02/2005 12:21:40	03/02/2005 12:21:40	0		Google Search: bird mating ca...	141	0	135

Figure 3.7: Firefox “history.dat” (3/4)

http://whyfiles.org/shorties/104chick_sev/	03/02/2005 12:21:56	03/02/2005 12:21:56	0	whyfiles.org	Chickadee Karaoke	142	0	42
http://www.google.co.uk/imgph?hl=en&tab=wi&...	03/02/2005 14:59:29	03/02/2005 14:59:29	0		Google Image Search	143	0	95
http://images.google.co.uk/images?client=Firefox...	03/02/2005 14:59:40	03/02/2005 14:59:40	0	images.google.co...	Google Search: young chicks	144	0	127
http://images.google.co.uk/imgres?imgurl=http://...	03/02/2005 14:59:56	03/02/2005 14:59:56	0		Google Image Result for http...	145	0	369
http://images.google.co.uk/imgres?imgurl=http://...	03/02/2005 14:59:56	03/02/2005 14:59:56	0			146	0	373
http://freespace.virgin.net/cobber/budgies/stories...	03/02/2005 14:59:56	03/02/2005 14:59:56	0	freespace.virgin.net	Scottie's Baby Pictures - 1	147	0	65
http://images.google.co.uk/imgres?imgurl=http://...	03/02/2005 15:01:02	03/02/2005 15:01:02	0		Google Image Result for http...	148	0	341
http://images.google.co.uk/imgres?imgurl=http://...	03/02/2005 15:01:02	03/02/2005 15:01:02	0			149	0	345
http://www.insaneanimals.com/funny-animals/17...	03/02/2005 15:01:08	03/02/2005 15:01:08	0	insaneanimals.com	Young Chicks - Funny Animal...	150	0	61
http://media.fastclick.net/w/get.media?tt=n&sid=...	03/02/2005 15:01:17	03/02/2005 15:01:17	0			151	0	503
http://pagead2.googlesyndication.com/pagead/a...	03/02/2005 15:01:17	03/02/2005 15:01:17	0			152	0	650
http://images.google.co.uk/images?q=young+chi...	03/02/2005 15:01:53	03/02/2005 15:01:53	0			153	0	122
http://images.google.co.uk/imgres?imgurl=http://...	03/02/2005 15:02:14	03/02/2005 15:02:14	0		Google Image Result for http...	154	0	383
http://images.google.co.uk/imgres?imgurl=http://...	03/02/2005 15:02:14	03/02/2005 15:02:14	0			155	0	386
http://www.cvm.okstate.edu/instruction/kocan/o...	03/02/2005 15:02:14	03/02/2005 15:02:14	0	cvm.okstate.edu		156	0	65
http://images.google.co.uk/images?q=young+chi...	03/02/2005 15:03:32	03/02/2005 15:03:32	0			157	0	122
http://images.google.co.uk/imgres?imgurl=http://...	03/02/2005 15:03:54	03/02/2005 15:03:54	0		Google Image Result for http...	158	0	346
http://images.google.co.uk/imgres?imgurl=http://...	03/02/2005 15:03:54	03/02/2005 15:03:54	0			159	0	349
http://people.cornell.edu/pages/sah67/summer.ht...	03/02/2005 15:03:54	03/02/2005 15:03:54	0	people.cornell.edu	Eco Gallery '04	160	0	49
http://www.google.co.uk/search?client=firefox-a...	09/02/2005 11:27:00	09/02/2005 11:27:00	0		Google Search: bird screensav...	161	0	135
http://www.traveltex.com/screen.asp?SNo=6245300...	09/02/2005 11:27:12	09/02/2005 11:27:12	0	traveltex.com	Screensavers	162	0	56
http://www.traveltex.com/download/screensaver...	09/02/2005 11:27:48	09/02/2005 11:27:48	0			163	0	57

Figure 3.8: Firefox “history.dat” (4/4)

BIRDWATCHING IN THAILAND
Feathered Magic from Mangroves to Mountaintops by Antony Lynam

One of the great attractions for nature tourists visiting Thailand is the diversity of environments extending across mountain peaks, lowland rainforests, mangroves, coral reefs, farmland and urban jungles. Within a day, or even a few hours travel, one can easily make the transition between these places and witness natural marvels large and small.

For wildlife enthusiasts no group maintains interest and pleasure more than birds. While special efforts are required to see in the wild charismatic species such as elephants and primates, birds are found across the entire spectrum of environments from pristine to severely degraded areas.

Part of the attraction for birds lies in their diversity. Nine hundred and seventy eight bird species have been recorded in Thailand, approximately 10% of the world's total. At the Isthmus of Kra between latitudes 11° and 13°N, a major biogeographic transition between Indochinese and Sundaic forests produces a special diversity of birds with a total of 152 species of birds reaching the northern or southern range limits of their geographic ranges.

Two-thirds of Thai birds are residents, the remainder are seasonal visitors. Locations where migrants congregate, often in large numbers, are highly accessible making Thailand a special destination for birdwatchers.

Many birds are susceptible to human disturbance because they have small geographic ranges, a result of specific habitat requirements. For example, Deignan's babbler - a non-descript forest bird is found on Doi Chiang Dao and nowhere else in the world. Gurney's pitta are only found in lowland rainforests. Fewer than 30 birds remain in the last known population in Thailand at Khao Nor Chuchi, Krabi. Efforts by local and international conservation agencies strive to increase protection efforts and reforest areas encroached by rubber farmers, though time is running out. With approximately 20% of the birds found in Thailand being globally or regionally endangered species, this makes the country a birdwatching haven for bird enthusiasts.

IDENTIFYING BIRDS

Birds are distinguished first by their size and shape. They range from diminutive flowerpeckers, sunbirds and white-eyes, about the size of your index finger, to lanky storks and egrets that stand almost a metre tall, and Green peafowl with its spectacular 2m tail. The form of the beak offers vital clues about the bird's diet. For example a thin curved tube for sipping nectar or a sharp hook for tearing

Figure 3.9: birdwatching.doc (1/9)

flesh. The pattern and colour of plumage can tell apart the sexes as in pheasants where males are bright and striking, and females are drab and dowdy. By far the most useful character for identification is a bird's voice. This is especially true in forests where on average 90% of birds are hidden from view. The most experienced birdwatchers in the tropics know their songs and calls.

BEST TIMES TO SEE BIRDS

The nesting season is a good time to be watching birds. During this most active time in a bird's life a variety of vocalizations and behaviours are exhibited. In Thailand, as in other tropical countries, the nesting season coincides with the period when food is in abundant supply. A bird expends much energy in courting, mating, incubating eggs, defending a nest and feeding offspring. Most birds nest during the transition between dry and wet seasons when new leaves and grass shoots sprout. This occurs from February to June. Certain birds depend on the availability of water and nest throughout the rainy season.

Migrants are best observed during their passage into or out of the country, or as they pass through on their way to other places. Most conspicuously, half a million ducks spend their winter in Thailand, feeding and resting in watery roosts from Chiang Saen to Thale Noi. Thousands of garganey and Lesser treeduck flock during January and February.

Shorebirds like sandpipers, stints and plovers migrate long-distances between nesting grounds in Eurasia and tropical Asia and wintering grounds in Australasia. They stop to feed in Thailand's mudflats and mangroves during September to May where they stock up on invertebrates and crustaceans. During October, the southward migration of hawks over peninsular Thailand is an avian spectacle. Chinese goshawks, Japanese sparrowhawks, crested honey buzzards, black bazas, and others are seen coasting on thermals in their thousands daily. Less conspicuous is the blue-winged pitta, a ground dwelling bird that arrives with the rains to nest in deciduous and bamboo forests, and escapes the hot season for the wetter forests of Malaysia and Sumatra.

Figure 3.10: birdwatching.doc (2/9)

WHERE TO FIND BIRDS

Given that many birds are denizens of certain times, places, habitats or seasons, the amateur naturalist can remember them by association.

PARKS, TEMPLES AND GARDENS

Some species like barn swallows, magpie robins, mynas and starlings can be found around Bangkok and environs. *Lumphini Park*, a heavily-used green area in the city centre supports a variety of birds with over 90 species having been recorded there. Temples near Bangkok and Ayutthaya preserve pockets of the natural landscape including birds such as black kites, parakeets and woodland birds that are characteristic of the habitats.

RICE PADDIES, MARSHES AND PONDS

Rice paddies, marshes and ponds away from built-up areas support breeding populations of Asian openbill stork and many other waterbirds.

Key sites: *Suphan Buri-Ayutthaya* and *Beung Boraphet*.

PEAT SWAMP FORESTS

Almost the last vestige of Thailand's peat swamp forest at *Phru To Daeng* or *Chalerm Phrakiat Wildlife Sanctuary* in Narathiwat supports Lesser adjutants, a kind of stork, along with several birds characteristic of Sundaic forests.

LAKES, RIVER SANDBANKS AND REEDBEDS

Lakes, river sandbanks and reedbeds preserve unique assemblages of wintering waterfowl and perching birds.

Key sites: *Chiang Saen*, *Fang Hot Springs*, and *Tha Ton*.

AGRICULTURAL LANDSCAPES

Agricultural landscapes across the country support species that tolerate human presence and include kites, rollers, bee-eaters, coucals, weavers and bulbuls.

SANDY BEACHES, MANGROVE AND TIDAL FLATS

Sandy beaches are attractive to tourists but are barren habitats for birds, while little-visited mangroves and tidal flats that are rich in nutrients and microorganisms, are favoured feeding haunts for migrant waders. Some birds like the Brown-winged kingfisher and Mangrove pitta, are entirely restricted to mangroves while Mangrove whistlers and flycatchers rarely leave the area.

Key sites: *Samut Sakhon*, *Ban Laem in Petchburi*, and *Krabi*.

OFFSHORE ISLANDS

Offshore islands such as *Phi Phi*, *Libong*, *Surin* and the *Similans* support fewer

Figure 3.11: birdwatching.doc (3/9)

species than similar sized mainland habitats but some such as Nicobar and Pied Imperial pigeon are entirely restricted to these refuges.

SEASCAPES

While there are fewer seabirds in the warm Thai waters compared with those in the northern and southern hemispheres, frigate birds, skuas, boobies, and terns are among the rewards for marine birders.

FORESTS

Most resident Thai birds depend upon forests for their survival. Rainforests in the extreme south support the greatest avian diversity, while seasonally dry dipterocarp, mixed deciduous and evergreen forests in the centre and north.

Key sites: *Khao Yai National Park*, *Kaeng Krachan National Park*, *Khao Soi Dao*, *Nam Nao National Park*, *Khao Nor Chuchi*, *Ban Nai Chong*, and *Hala Bala Wildlife Sanctuary*.

MOUNTAINS

Sibias, minlars, and laughing thrushes are relatives of species found in the Himalayas and southern China, and can only be found in mountain forests. Following surveys of high mountain peaks in the last decade, at least 20 new species or 2% of the total have been added to the lists for Thailand.

Key sites: *Doi Pha Hom Pok*, *Doi Chiang Dao*, *Doi Inthanon*, *Doi Ang Khang*, and *Doi Suthep*.

By visiting these enchanting destinations, travellers can appreciate the avian wonders that Thailand offers and better understand the importance of the wild and not so wild areas that preserve them.

Contact information:

Bird Conservation Society of Thailand (BCST)*
69/12 Soi Ramindra 24, Jorakaebau, Ladprao, Bangkok 10230, Thailand
Email: bcst@box1.a-net.net.th
Tel: 66-(0)-2943-5965

Web sites:
www.bcst.org/index_ebird.html
www.thai.net/bcst

* The Bird Conservation Society of Thailand (BCST) is a BirdLife Partner

Oriental Bird Club (OBC)
c/o Uthai Treesucon, 723/1 Mu 2 Soi Ram Intra, Jorakhaebau, Bangkok 10230.
E-mail:
utree@loxinfo.co.th
mail@orientalbirdclub.org
Web site: www.orientalbirdclub.org

Figure 3.12: birdwatching.doc (4/9)

Wildlife Conservation Society - Thailand Programme
P.O. Box 170, Laksi, Bangkok 10210
Tel: +662-503 4478, +662-503 4479
Fax: +662-503 4096
Email: thailand@wcs.org

Reference information:

Field guide
Robson, C. 2002.
A field guide to the birds of Thailand.
Asia Books, Bangkok. 272pp.

TAT PUBLICATIONS ON NATURE TOURISM

For more information on birdwatching in Thailand, please refer to the following nature tourism guide books in the "National Park" series published by the Tourism Authority of Thailand as part of the Tourism and Employment Creation Plan implemented under the Social Investment Project.

- KHAO YAI: DONG PHAYA YEN FOREST RANGE
ISBN 974-8252-70-1
- KAENG KRACHAN:
Amazing Forest of Phetchburi River
ISBN 974-8252-72-8
- KHAO SOK
Fascinating Limestone Mountains Amid The Verdant Forest of Surat Thani,
Thailand
ISBN 974-679-099-4
- DOI INTANON - DOI SUTHEP
The Himalayan Springs of Thailand
SBN 974-8252-67-1
- PHU LUANG
The Window of Nature

Figure 3.14: birdwatching.doc (5/9)

The Kingdom of Plants
ISBN 974-8252-73-6

BIRDWATCHING TIPS

- Consult field guides, checklists, and maps prior to arriving at the birdwatching destination.
- Dress in colours that blend in with the surroundings.
- Bring appropriate equipment such as telescope, bird guide, and a notebook.
- Plan to arrive at the destination at sunrise when birds are first out in search for food and are most active.
- Walk slowly stopping at intervals to listen for calling birds.
- Look for the birds in thickets, on the branches of trees, and on the ground.
- Avoid talking, smoking, or walking on dry leaves, all of which will scare birds away.

CONSERVING BIRD DIVERSITY

Across the world, humans coexist with birds but human activities often affect the survival prospects for birds. Forty-eight species of birds found in Thailand (5% of the total) are globally threatened by the loss or disturbance of their habitats, food resources, and breeding areas, and by other human actions, and so require special conservation attention. A further 97 species (10%) are potentially threatened if current trends persist.

In Thailand, 101 species (10% of the total) are hunted as pests, for food or for the pet trade and are directly threatened by humans. The list of hunted species includes waterbirds, birds of prey, pheasants, parakeets, pigeons, hornbills, pittas weavers, bulbuls, and other forest birds. Worldwide the trade in birds includes 2,600 species and several million birds each year. Ten percent of threatened birds worldwide are affected by the bird trade.

There are a number of ways in which the numbers and distribution of threatened birds are being restored. Habitat conservation programmes are an important mechanism. Feeding habitats for seasonal migrants are being preserved by incorporating mangroves and coastal tidal areas in marine protected areas. Lowland forests that support Gurney's pitta and other diversity are being protected and a reforestation programme planned. Preventing encroachment around the edges of parks maintains the integrity of forest blocks used by the majority of native birds, including migrant raptors and songbirds.

Figure 3.15: birdwatching.doc (6/9)

In most cases, preserving and protecting natural habitats can bring back even highly endangered populations of rare birds. These measures are relatively cost effective to implement. As an example, a 3-year Khao Yai Conservation Project preserved over 2,000 sq km of wildlife habitat, discouraged poaching, and provided employment alternatives for local forest resource users, at an annual cost of 7 million baht (US\$171,000).

For critically endangered species, whose populations are extremely small, and whose survival in the wild is uncertain due to factors that simply cannot be controlled, other more costly measures such as captive breeding, are being considered. Captive breeding is risky because birds need to be recovered from the wild to establish breeding populations, and injury is possible. Some species do not breed well in captivity because their natural courting and nesting behaviours are no longer possible. Strict controls on who is allowed to breed endangered species, registration of individuals, and enforcement of laws so that commercial sale is not possible, need to be adopted. Without these controls, captive breeding programmes cannot succeed.

HOW YOU CAN HELP IN THE CONSERVATION OF BIRDS

Visitors to Thailand can assist efforts to preserve and maintain the diversity of birds and their habitats simply by visiting national parks and other wilderness areas. Bird enthusiasts can report the species they observe to authorities. Checklists are now available at many popular national parks. Tourists can report evidence of suspicious activity that might lead to the arrest of unscrupulous individuals trapping or hunting birds and can also participate as volunteers in habitat conservation programmes. In these ways, tourists can help reduce the threats to birds, and at the same time enjoy Thailand's birdwatching paradise.

Contact information:

TO REPORT BIRD SPECIES OBSERVED

- **TO REPORT BIRD SPECIES OBSERVED**
Please contact the Park Visitor Centre of the National Park
or
Bird Conservation Society of Thailand (BCST)*
69/12 Soi Ramindra 24, Jorakaebau, Ladprao, Bangkok 10230, Thailand
Email: bcst@box1.a-net.net.th
Tel: 66-(0)-2943-5965
Web sites:
www.bcst.org/index_ebird.html
www.thai.net/bcst
* The Bird Conservation Society of Thailand (BCST) is a BirdLife Partner

- **TO REPORT EVIDENCE OF SUSPICIOUS ACTIVITY**

Figure 3.16: birdwatching.doc (7/9)

Please contact Wildlife Protection and Suppression Office Department of National Parks, Wildlife, and Plant Conservation 61 Paholyothin Road, Chatuchak, Bangkok 10900 Tel: 66-(0)-2579-5266
• HABITAT CONSERVATION VOLUNTEER PROGRAMMES Please contact Wildlife Conservation Society - Thailand Programme P.O. Box 170, Laksi, Bangkok 10210
Tel: +662-503 4478, +662-503 4479 Fax: +662-503 4096 Email: thailand@wcs.org FAMILIES OF BIRDS IN THAILAND UNDER THREAT The following bird species are under threat because there is a high demand for them and they are hunted for the local, regional and global bird trade. To help preserve the species, please refrain from purchasing any of the following birds as pets, and if you happen to witness any of the following birds being sold or traded, or note any suspicious activities involving them, please contact: The Wildlife Protection and Suppression Office Department of National Parks, Wildlife, and Plant Conservation 61 Paholyothin Road, Chatuchak, Bangkok 10900 Tel: 66-(0)-2579-5266 1. Phasianidae (wood partridges and pheasants) - 4 species 2. Anatidae (White-winged duck) - 1 species 3. Picidae (woodpeckers and barbets) - 3 species 4. Bucerotidae (hornbills) - 7 species 5. Upupidae (Common hoopoe) - 1 species 6. Cuculidae (Coral-billed ground cuckoo) - 1 species

Figure 3.17: birdwatching.doc (8/9)

6. Cuculidae (Coral-billed ground cuckoo) - 1 species
7. Psittacidae (parrots and parakeets) - 4 species
8. Columbidae (pigeons) - 12 species
9. Accipitridae (birds of prey) - 9 species
10. Threskiornithidae (White-shouldered ibis) - 1 species
11. Ciconiidae (Lesser adjutant) - 1 species
12. Pittidae (pittas) - 3 species
13. Irenidae (Asian fairy bluebird and leafbirds) - 6 species
14. Corvidae (jays, crows, magpies, orioles and minivets) - 14 species
15. Muscicapidae (thrushes, robins, and sharmas) - 3 species
16. Sturnidae (starlings and mynas) - 3 species
17. Paridae (Yellow-cheeked tit) - 1 species
18. Pycnonotidae (bulbuls) - 7 species
19. Zosteropidae (Japanese white-eye) - 1 species
20. Sylviidae (laughing thrushes, mesias, minlas and sibias) - 8 species
21. Nectariniidae (Scarlet-backed flowerpecker) - 1 species
22. Passeridae (weavers and munias) - 7 species
23. Fringillidae (grosbeaks and buntings) - 3 species

About The Author

ANTONY LYNAM

Antony Lynam (Ph.D.), Wildlife Conservation Society (WCS)- Thailand Programme Director and conservation scientist, works with the Thailand Department of National Parks, Wildlife and Plants to develop programmes for the conservation of the country's endangered species, park resources management, and the design and conduct of training curriculum for park rangers.

An Australian citizen, he has authored a number of technical papers and popular articles concerning conservation issues in Australia, North America, and Thailand, and was a contributor to the seminal volume on habitat fragmentation "Tropical Forest Remnants: Ecology, Conservation and Management". He writes frequently on natural history for magazines, journals and newspapers including Wildlife Conservation, The Nation, The Bangkok Post, and The Natural History Bulletin of The Siam Society.

Figure 3.18: birdwatching.doc (9/9)

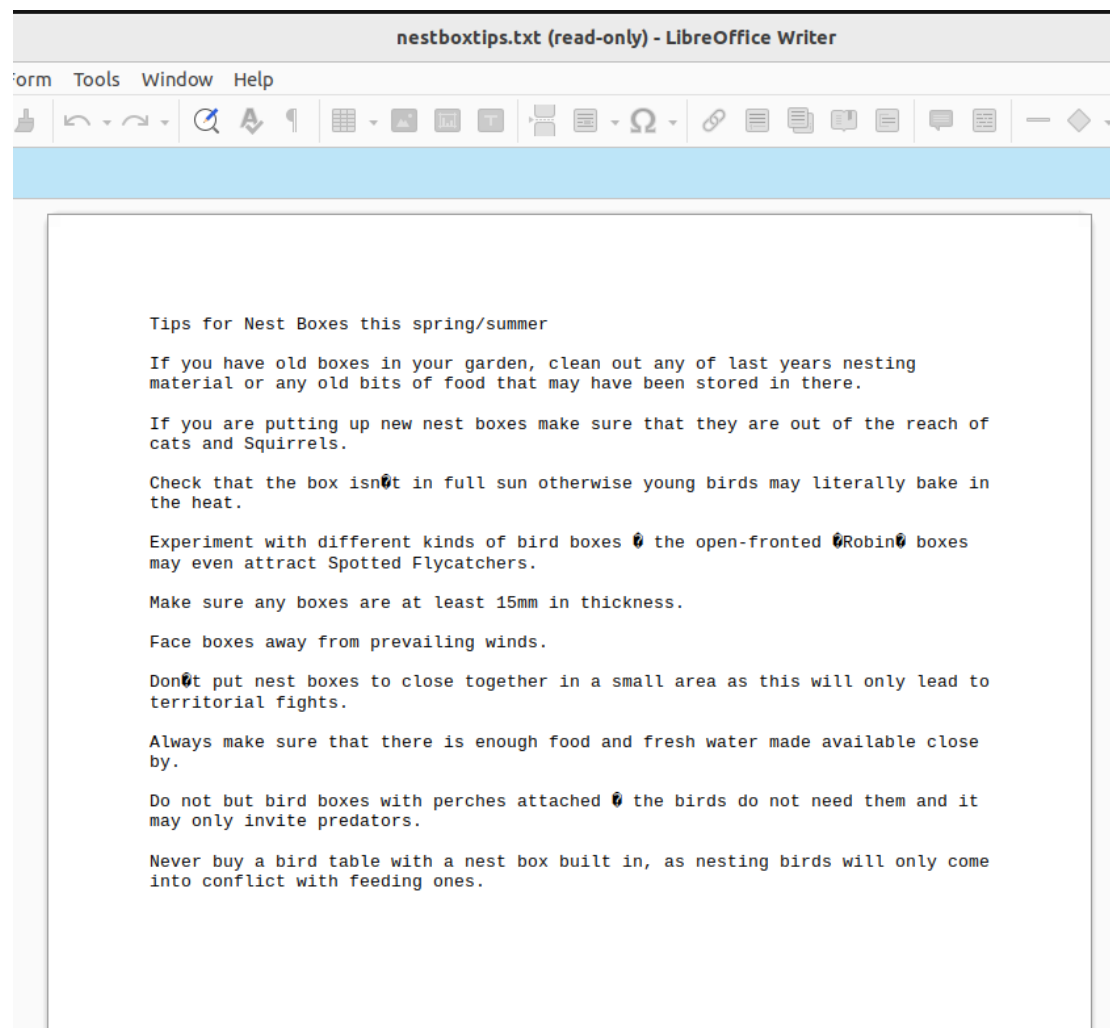


Figure 3.19: nestboxtips.txt

RID	Username	Admin?	Lock?
01f4	Administrator	ADMIN	
03ed	bob		
01f5	Guest		dis/lock
03e8	HelpAssistant		dis/lock
03ec	jane		
03eb	johndoe	ADMIN	
03ea	SUPPORT_388945a0		dis/lock

File Edit Reports Help

Node	Last mod. time
AcroRd32.exe	2005/02/02 16:59:47
bckgzm.exe	2005/01/24 15:36:16
chkrzm.exe	2005/01/24 15:36:16
cmimg32.exe	2005/01/24 15:32:42
CONF.EXE	2005/01/24 15:39:40
dialer.exe	2005/01/24 15:36:15
excel.exe	2005/01/25 10:49:38
Firefox.exe	2005/01/24 16:02:43
HELPCTR.EXE	2005/01/24 15:39:50
hrtzmm.exe	2005/01/24 15:36:16
hypertrm.exe	2005/01/24 15:36:16
ICWCONN1.EXE	2005/01/24 15:39:39
ICWCONN2.EXE	2005/01/24 15:39:39
IEKPLOR.E	2005/01/24 15:39:29
INETWIZ.EXE	2005/01/24 15:39:39
infopath.exe	2005/01/25 10:49:45
install.exe	2005/01/24 15:40:04
ISGNUMP.EXE	2005/01/24 15:39:39
milgwis.exe	2005/01/24 16:12:35
moviemk.exe	2005/01/24 15:39:51
player2.exe	2005/01/24 15:39:42
MSCONFIG.EXE	2005/01/24 15:39:50
mslsm.exe	2005/01/24 15:43:16
msinfo32.exe	2005/01/24 15:39:40
MSMSSCS.EXE	2005/01/24 15:35:33
MsoHtmEd.exe	2005/01/25 10:49:42
msxmlmed.exe	2005/01/25 10:49:45
MSPLUB.EXE	2005/01/25 10:49:43
OUTLOOK.EXE	2005/01/25 10:49:43
pbrush.exe	2005/01/24 15:40:04
pinball.exe	2005/01/24 15:36:16
powerpnt.exe	2005/01/25 10:49:43
RealPlay.exe	2005/02/02 14:50:14
rmproc.exe	2005/02/02 14:49:59
rsnsm.exe	2005/01/24 15:36:16
setup.exe	2005/01/24 15:40:04
shvzmm.exe	2005/01/24 15:36:16
table30.exe	2005/01/24 15:40:04
thunderbird.exe	2005/01/24 16:31:47
wab.exe	2005/01/24 15:39:40
wabmig.exe	2005/01/24 15:39:40
winnl32.exe	2005/01/24 15:40:04
Winword.exe	2005/01/25 10:49:44
wmplayer.exe	2005/01/24 15:39:55
WORDPAD.EXE	2005/01/24 15:36:15
WRITE.EXE	2005/01/24 15:36:15

Forensic Registry Editor (fred) v0.2.0 - /home/cmp209/jd/registry/software (read-only)

Key	Type	Value
LocalNt\LOCAL_SZ	REG_SZ	C:\Program Files\Mozilla Thunderbird\thunderbird.exe
Path	REG_SZ	C:\Program Files\Mozilla Thunderbird\

Hex viewer

0000	0100	0200	0300	0400	0500	0600	0700	0800	0900	0A00	0B00	0C00	0D00	0E00	0F00	Comment	Int8
61	60	68	60	20	00	46	69	60	60	67	60	72	00	00	00	C:\P.r.o.g.r.a.m.F.i.l.e.s.	int8: 67
0020	5c	00	69	00	60	60	60	60	60	61	00	00	00	00	00	\\M.o.s.i.l.l.a.	int8: 67
0030	20	00	54	60	68	00	75	00	60	64	00	65	00	72	00	.T.h.u.n.d.e.r.	int16: 67
0040	62	00	64	60	72	64	00	5c	00	74	00	68	00	75	00	b.i.r.d.v.t.h.u.	int32: 3801155
0050	60	60	64	00	65	00	72	00	62	69	00	72	00	64	00	n.d.e.r.b.i.r.d.	int32: 3801155
0060	2e	00	05	60	78	00	65	00	00	00	00	00	00	00	00	..e.x.e..	unixtime: 1970/02/13 23:52:35

Byte offset: 0x0000 (0)

int64	int32	int16	float	double
int64:	22518393277644867			
int32:	22518393277644867			
int16:	22518393277644867			
float:	22518393277644867			
double:	22518393277644867			

The screenshot shows the Forensic Registry Editor (fred) v0.2.0 interface. The title bar indicates the file path: /home/cmp209/jd/registry/system (read-only). The menu bar includes File, Edit, Reports, and Help. The left pane shows a tree view of the registry structure, with the 'MountedDevices' key selected. The right pane displays the hex view of the selected value, showing the raw data in hexadecimal and its ASCII representation.

Node	Last mod. time	Key	Type	Value
ControlSet001	2005/02/02 12:35:26	\\?\\Volume{30bf5ac0-6e1f-11d9-a7bd-806d617269f}	REG_BINARY	5c 00 3f 00 3f 00 5c 00 46 00 44 00 43 00 23 00 47 00 45 00 4e 00 45 00
ControlSet002	2005/01/24 15:53:25	\\?\\Volume{30bf5ac1-6e1f-11d9-a7bd-806d617269f}	REG_BINARY	5c 00 3f 00 3f 00 5c 00 49 00 44 00 45 00 23 00 43 00 64 00 52 00 6f 00
LastKnownGoodRecovery	2005/02/09 11:03:35	\\?\\Volume{30bf5ac3-6e1f-11d9-a7bd-806d617269f}	REG_BINARY	b8 dd b8 dd 00 7e 00 00 00 00 00 00
MountedDevices	2005/02/03 11:18:01	\\?\\Volume{44d36d3b-7525-11d9-ab5a-0048545652e0}	REG_BINARY	5c 00 3f 00 3f 00 5c 00 53 00 54 00 4f 00 52 00 41 00 47 00 45 00 23 00
Select	2005/01/24 15:53:27	\\?\\Volume{44d36d3b-7525-11d9-ab5a-0048545652e0}	REG_BINARY	b8 dd b8 dd 00 fc 47 bb 00 00 00 00
Setup	2005/01/24 15:53:33	\\DosDevices\\A:	REG_BINARY	5c 00 3f 00 3f 00 5c 00 46 00 44 00 43 00 23 00 47 00 45 00 4e 00 45 00
WPA	2005/01/24 15:57:00	\\DosDevices\\C:	REG_BINARY	b8 dd b8 dd 00 7e 00 00 00 00 00 00
		\\DosDevices\\D:	REG_BINARY	5c 00 3f 00 3f 00 5c 00 49 00 44 00 45 00 23 00 43 00 64 00 52 00 6f 00
		\\DosDevices\\E:	REG_BINARY	5c 00 3f 00 3f 00 5c 00 53 00 54 00 4f 00 52 00 41 00 47 00 45 00 23 00
		\\DosDevices\\F:	REG_BINARY	b8 dd b8 dd 00 fc 47 bb 00 00 00 00

Hex viewer

0000	5c 00 3f 00 3f 00 5c 00 53 00 54 00 4f 00 52 00	\\?.?.\S.T.O.R.	int8:	92
0010	41 00 47 00 45 00 23 00 52 00 65 00 6d 00 6f 00	A.G.E.#.R.e.m.o.	uint8:	92
0020	76 00 61 00 62 00 6c 00 65 00 4d 00 65 00 64 00	v.a.b.l.e.M.e.d.	int16:	92
0030	69 00 61 00 23 00 37 00 60 00 32 00 66 00 62 00	1.a.#.7.6.2.f.b.	uint16:	92
0040	34 00 32 00 37 00 64 00 63 00 26 00 30 00 26 00	4.2.7.d.c.6.0.6.	int32:	4128860
0050	52 00 4d 00 23 00 7b 00 35 00 33 00 66 00 35 00	R.M.#.{.5.3.f.5.	uint32:	4128860
0060	36 00 33 00 30 00 64 00 2d 00 62 00 36 00 62 00	6.3.0.d.-.b.6.b.	uintmax:	1970/02/17 18:54:20
0070	66 00 2d 00 31 00 31 00 64 00 30 00 2d 00 39 00	f.-.1.1.d.0.-.9.	int64:	25895968444448860
0080	34 00 66 00 32 00 2d 00 30 00 30 00 61 00 30 00	4.f.2.-.0.0.a.0.	uint64:	25895968444448860
0090	63 00 39 00 31 00 65 00 66 00 62 00 38 00 62 00	c.9.1.e.f.b.8.b.	filetime64:	2091/05/14 23:52:13
00a0	7d 00	},		

Byte offset: 0x0000 (0)

• Little endian ○ Big endian

Figure 4.2.1: Removable Media E

File	Edit	Reports	Help
Node	Last mod. time	Key	Type Value
ControlSet001	2005/02/02 12:35:26	\\?Volume{44d36d3b-7525-11d9-ab5a-0048545652e0}	REG_BINARY 5c 00 3f 00 3f 00 5c 00 53 00 54 00 4f 00 52 00 41 00 47 00 45 00 23 00
ControlSet002	2005/01/24 15:53:25	\\?Volume{44d36d43-7525-11d9-ab5a-0048545652e0}	REG_BINARY b8 dd b8 dd 00 fc 47 bb 00 00 00 00
LastKnownGoodRecovery	2005/02/09 11:03:35	\\DosDevices\\A:	REG_BINARY 5c 00 3f 00 3f 00 5c 00 46 00 44 00 43 00 23 00 47 00 45 00 4e 00 45 00
MountedDevices	2005/02/03 11:30:32	\\DosDevices\\C:	REG_BINARY b8 dd b8 dd 00 7e 00 00 00 00 00 00
Select	2005/01/24 15:53:27	\\DosDevices\\D:	REG_BINARY 5c 00 3f 00 3f 00 5c 00 49 00 44 00 45 00 23 00 43 00 64 00 52 00 6f 00
Setup	2005/01/24 15:53:33	\\DosDevices\\E:	REG_BINARY 5c 00 3f 00 3f 00 5c 00 53 00 54 00 4f 00 52 00 41 00 47 00 45 00 23 00
WPA	2005/01/24 15:57:00	\\DosDevices\\F:	REG_BINARY b8 dd b8 dd 00 fc 47 bb 00 00 00 00

Hex viewer			
0000 b8 dd b8 dd 00 fc 47 bb 00 00 00 00	G.....	int8: -72	
		uint8: 184	
		int16: -8776	
		uint16: 56760	
		int32: -575087176	
		uint32: 3719880120	
		unixtime: 2087/1/17 04:02:00	
		int64: -4951712184620491336	
		uint64: 13495031889089060280	
		filetime64: 2105/10/19 17:22:42	

Figure 4.2.2: Hidden F Partition

Node	Last mod. time	Key	Type Value
ControlSet001	2005/02/02 12:35:26	Capabilities	REG_DWORD 0x00000010
ControlSet002	2005/02/09 03:10:25	Class	REG_SZ DiskDrive
Enum	2005/02/02 16:29:57	ClassGUID	REG_SZ {4D36E967-E325-11CE-BFC1-08002BE10318}
ACPI	2005/01/24 16:09:03	CompatibleIDs	REG_MULTI_SZ USBSTOR\\Disk USBSTOR\\RAW
ACPI_HAL	2005/01/28 15:50:16	ConfigFlags	REG_DWORD 0x00000000
FDC	2005/01/24 16:09:03	DeviceDesc	REG_SZ Disk drive
HTRFEE	2005/01/24 16:09:03	Driver	REG_SZ {4D36E967-E325-11CE-BFC1-08002BE10318}\\0002
IDE	2005/01/28 15:50:02	FriendlyName	REG_SZ USB DISK USB Device
ISAPNP	2005/01/24 16:09:03	HardwareID	REG_MULTI_SZ USBSTOR\\Disk USB_DISK_1.05 USBSTOR\\Disk USB_DISK_ USBSTOR\\Disk USB_DISK_1
LPTENUM	2005/01/24 16:15:18	Mfg	REG_SZ (Standard disk drives)
PCI	2005/01/24 16:09:03	ParentIdPrefix	REG_SZ 7&2fb427dc&0
PCIIDE	2005/02/09 03:10:17	Service	REG_SZ disk
Root	2005/02/02 16:29:57	UINumber	REG_DWORD 0x00000000
STORAGE	2005/02/02 16:29:57		
SW	2005/02/02 16:29:57		
USB	2005/02/02 16:29:57		
USBSTOR	2005/02/02 16:29:57		
Disk&Ven&Prod USB DISK&Rev 1.05	2005/02/02 16:29:57		
Device Parameters	2005/02/02 16:30:06		
MediaChangeNotification	2005/02/02 16:29:57		
LogConf	2005/02/09 03:09:53		
Hardware Profiles	2005/02/09 03:10:38		
Services	2005/02/02 16:29:57		
ControlSet002	2005/02/09 03:10:25		
Enum	2005/02/02 16:29:57		

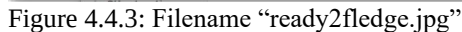
Figure 4.2.3: Past connected USB

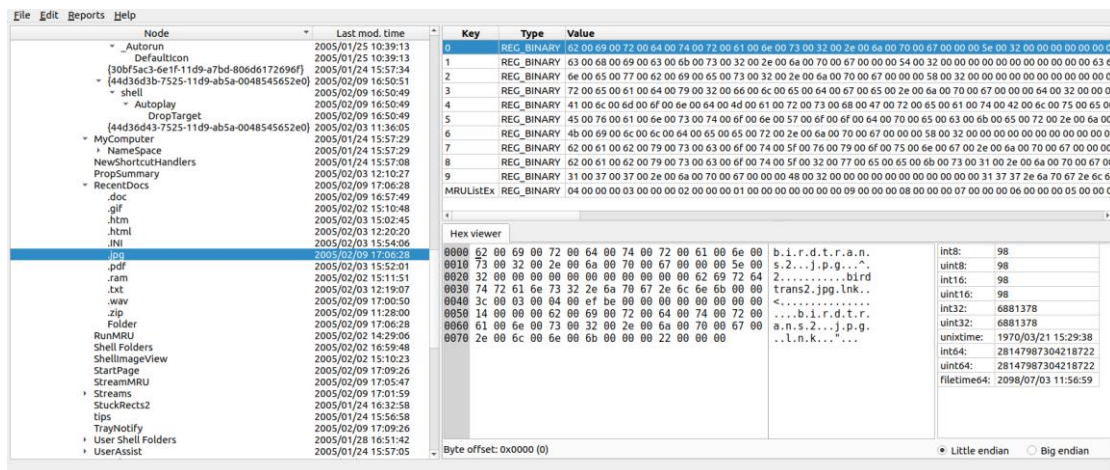
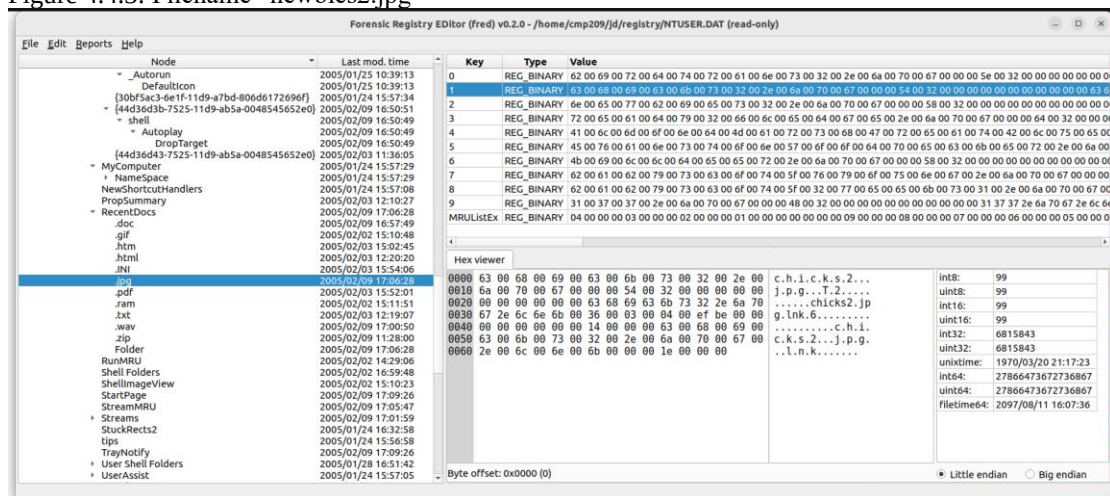
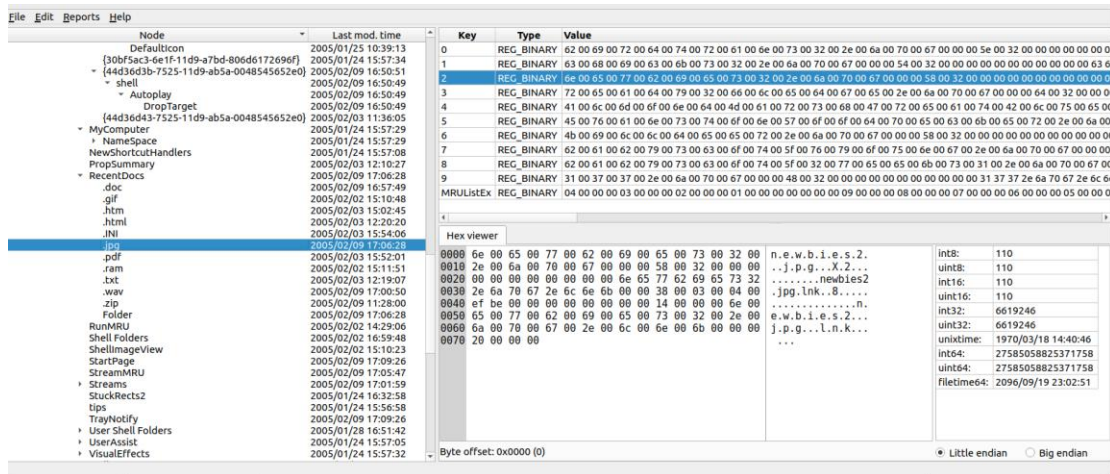
Node	Last mod. time	Key	Type Value
ControlSet001	2005/02/02 12:35:26	Capabilities	REG_DWORD 0x00000010
ControlSet002	2005/02/09 03:10:25	Class	REG_SZ DiskDrive
Enum	2005/02/02 16:29:57	ClassGUID	REG_SZ {4D36E967-E325-11CE-BFC1-08002BE10318}
ACPI	2005/01/24 16:09:03	CompatibleIDs	REG_MULTI_SZ USBSTOR\\Disk USBSTOR\\RAW
ACPI_HAL	2005/01/28 15:50:16	ConfigFlags	REG_DWORD 0x00000000
FDC	2005/01/24 16:09:03	DeviceDesc	REG_SZ Disk drive
HTRFEE	2005/01/24 16:09:03	Driver	REG_SZ {4D36E967-E325-11CE-BFC1-08002BE10318}\\0002
IDE	2005/01/28 15:50:02	FriendlyName	REG_SZ USB DISK USB Device
ISAPNP	2005/01/24 16:09:03	HardwareID	REG_MULTI_SZ USBSTOR\\Disk USB_DISK_1.05 USBSTOR\\Disk USB_DISK_ USBSTOR\\Disk USB_DISK_1
LPTENUM	2005/01/24 16:15:18	Mfg	REG_SZ (Standard disk drives)
PCI	2005/01/24 16:09:03	ParentIdPrefix	REG_SZ 7&2fb427dc&0
PCIIDE	2005/02/09 03:10:17	Service	REG_SZ disk
Root	2005/02/02 16:29:57	UINumber	REG_DWORD 0x00000000
STORAGE	2005/02/02 16:29:57		
SW	2005/02/02 16:29:57		
USB	2005/02/02 16:29:57		
USBSTOR	2005/02/02 16:29:57		
Disk&Ven&Prod USB DISK&Rev 1.05	2005/02/02 16:29:57		
Device Parameters	2005/02/02 16:30:06		
MediaChangeNotification	2005/02/02 16:29:57		
LogConf	2005/02/09 03:09:53		
Hardware Profiles	2005/02/09 03:10:38		
Services	2005/02/02 16:29:57		
ControlSet002	2005/02/09 03:10:25		
Enum	2005/02/02 16:29:57		

Figure 4.2.4: Mounted Devices

Node	Last mod. time	Key	Type Value
ControlSet001	2005/02/02 12:35:26	Capabilities	REG_DWORD 0x00000010
ControlSet002	2005/02/09 03:10:25	Class	REG_SZ DiskDrive
Enum	2005/02/02 16:29:57	ClassGUID	REG_SZ {4D36E967-E325-11CE-BFC1-08002BE10318}
ACPI	2005/01/24 16:09:03	CompatibleIDs	REG_MULTI_SZ USBSTOR\\Disk USBSTOR\\RAW
ACPI_HAL	2005/01/28 15:50:16	ConfigFlags	REG_DWORD 0x00000000
FDC	2005/01/24 16:09:03	DeviceDesc	REG_SZ Disk drive
HTRFEE	2005/01/24 16:09:03	Driver	REG_SZ {4D36E967-E325-11CE-BFC1-08002BE10318}\\0002
IDE	2005/01/28 15:50:02	FriendlyName	REG_SZ USB DISK USB Device
ISAPNP	2005/01/24 16:09:03	HardwareID	REG_MULTI_SZ USBSTOR\\Disk USB_DISK_1.05 USBSTOR\\Disk USB_DISK_ USBSTOR\\Disk USB_DISK_1
LPTENUM	2005/01/24 16:15:18	Mfg	REG_SZ (Standard disk drives)
PCI	2005/01/24 16:09:03	ParentIdPrefix	REG_SZ 7&2fb427dc&0
PCIIDE	2005/02/09 03:10:17	Service	REG_SZ disk
Root	2005/02/02 16:29:57	UINumber	REG_DWORD 0x00000000
STORAGE	2005/02/02 16:29:57		
SW	2005/02/02 16:29:57		
USB	2005/02/02 16:29:57		
USBSTOR	2005/02/02 16:29:57		
Disk&Ven&Prod USB DISK&Rev 1.05	2005/02/02 16:29:57		
Device Parameters	2005/02/02 16:30:06		
MediaChangeNotification	2005/02/02 16:29:57		
LogConf	2005/02/09 03:09:53		
Hardware Profiles	2005/02/09 03:10:38		
Services	2005/02/02 16:29:57		
ControlSet002	2005/02/09 03:10:25		
Enum	2005/02/02 16:29:57		

Figure 4.3: Computer description file showing connections to external computers





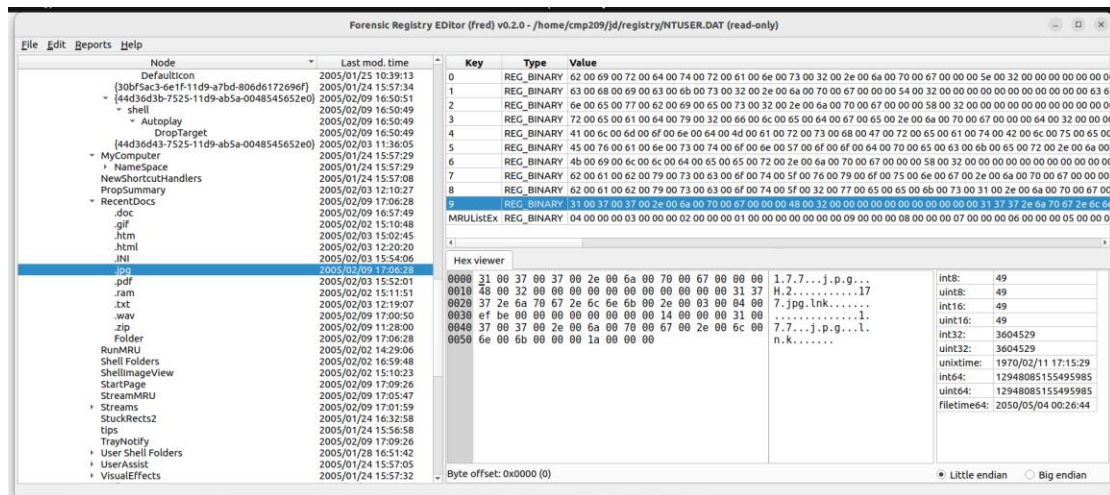


Figure 4.4.6 Filename “177.jpg”

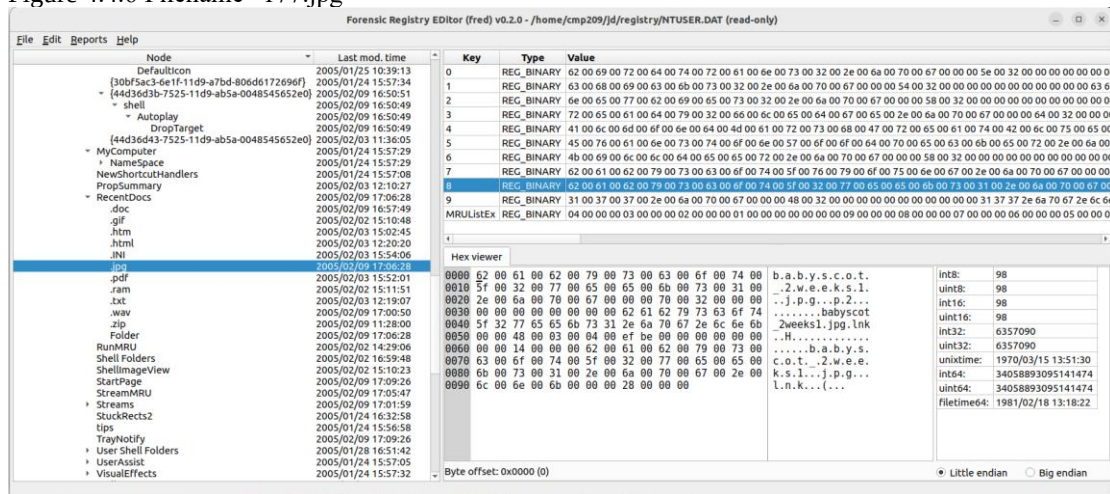


Figure 4.4.7 Filename “babyscot_2weeks1.jpg”

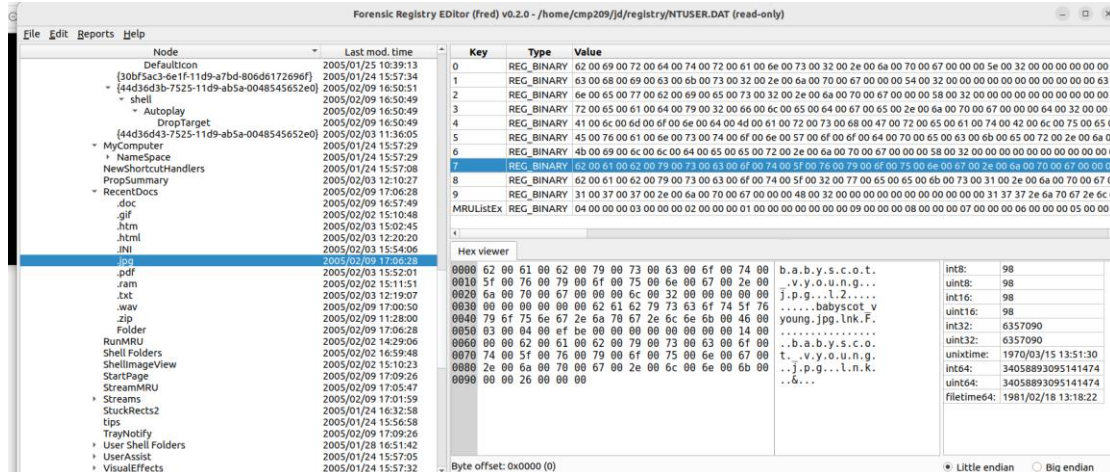


Figure 4.4.8: Filename “babyscot_vyoung.jpg”

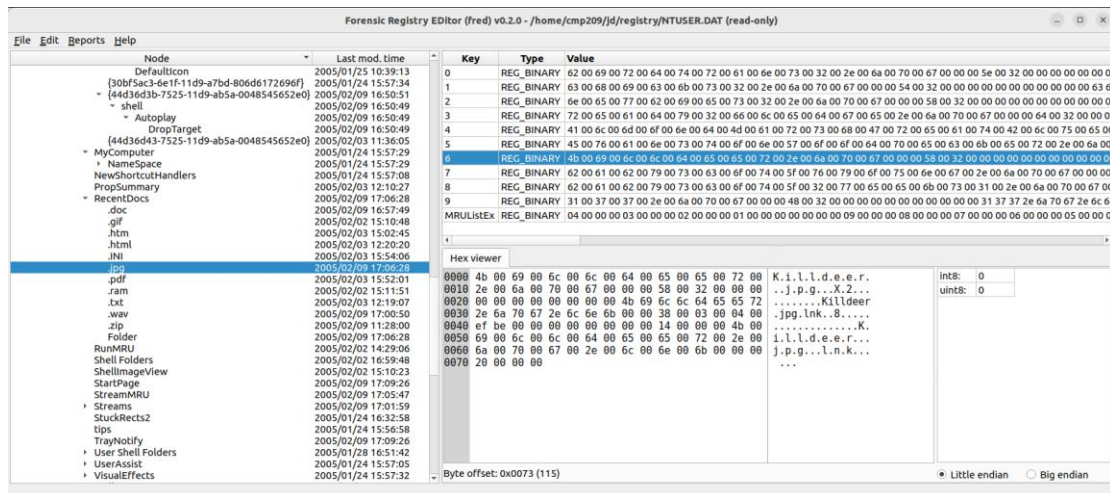


Figure 4.4.9: Filename “Killdeer.jpg”

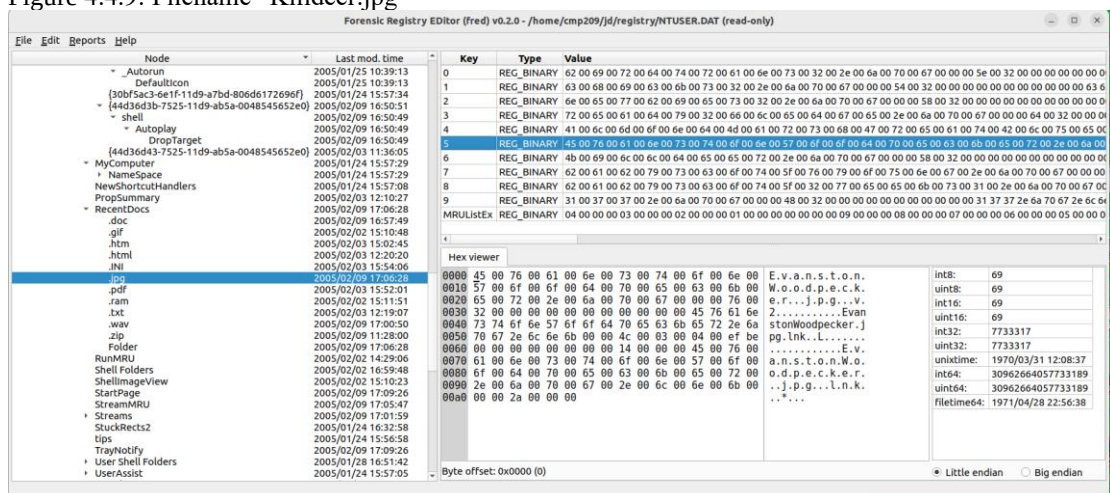


Figure 4.4.10: Filename “EvanstonWoodpecker.jpg”

```

cmp209@cmp209-virtual-machine:~/suspectDrive$ find . -name "AlmondMarshGreatBlueHeronStalling.jpg" -print
cmp209@cmp209-virtual-machine:~/suspectDrive$ find . -name "ready2fledge.jpg" -print
./Documents and Settings/bob/My Documents/My Music/ready2fledge.jpg
cmp209@cmp209-virtual-machine:~/suspectDrive$ find . -name "newbies2.jpg" -print
./Documents and Settings/johndoe/My Documents/newbies2.jpg
cmp209@cmp209-virtual-machine:~/suspectDrive$ find . -name "chicks2.jpg" -print
./Documents and Settings/johndoe/My Documents/My Pictures/chicks2.jpg
cmp209@cmp209-virtual-machine:~/suspectDrive$ find . -name "birdtrans2.jpg" -print
./Documents and Settings/johndoe/Desktop/birdtrans2.jpg
cmp209@cmp209-virtual-machine:~/suspectDrive$ find . -name "177.jpg" -print
./Documents and Settings/johndoe/My Documents/My Pictures/177.jpg
cmp209@cmp209-virtual-machine:~/suspectDrive$ find . -name "babyscot_2weeks1.jpg" -print
./Documents and Settings/johndoe/My Documents/My Pictures/babyscot_2weeks1.jpg
cmp209@cmp209-virtual-machine:~/suspectDrive$ find . -name "babyscot_vyoung.jpg" -print
./Documents and Settings/johndoe/My Documents/My Pictures/babyscot_vyoung.jpg
cmp209@cmp209-virtual-machine:~/suspectDrive$ find . -name "killdeer.jpg" -print
cmp209@cmp209-virtual-machine:~/suspectDrive$ find . -name "Killdeer.jpg" -print
cmp209@cmp209-virtual-machine:~/suspectDrive$ find . -name "EvanstonWoodpecker.jpg" -print

```

Figure 4.4.11: Filename follow up find

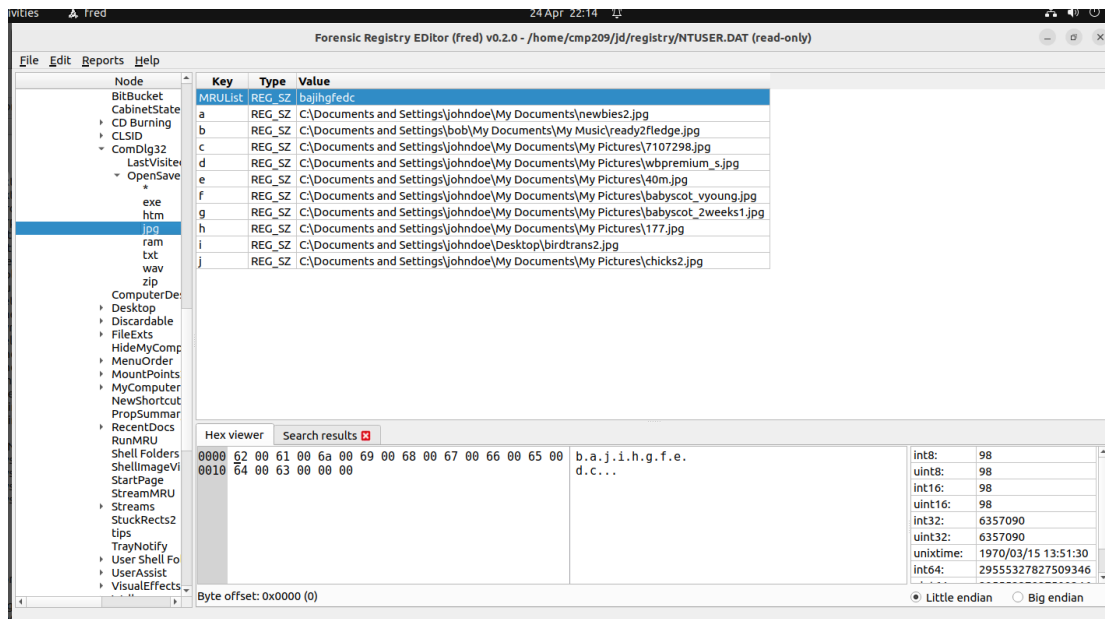


Figure 4.4.12: Filenames and locations in OpenSaveMRU

Metadata

Name: .jpg

Number of subkeys: 0

Number of values: 11

Modification Time: 2005-02-09 17:06:28 GMT+00:00

Name	Type	Value
0	REG_BIN	62 00 69 00 72 00 64 00 74 00 72 00 61 00 6E 00...
MRUListEx	REG_BIN	04 00 00 00 03 00 00 00 02 00 00 00 01 00 00 00...
1	REG_BIN	63 00 68 00 69 00 63 00 6B 00 73 00 32 00 2E 00...
2	REG_BIN	6E 00 65 00 77 00 62 00 69 00 65 00 73 00 32 00...
3	REG_BIN	72 00 65 00 61 00 64 00 79 00 32 00 66 00 6C 00...
4	REG_BIN	41 00 6C 00 6D 00 6F 00 6E 00 64 00 4D 00 61 00...
5	REG_BIN	45 00 76 00 61 00 6E 00 73 00 74 00 6F 00 6E 00...
6	REG_BIN	4B 00 69 00 6C 00 6C 00 64 00 65 00 65 00 72 00...
7	REG_BIN	62 00 61 00 62 00 79 00 73 00 63 00 6F 00 74 00...
8	REG_BIN	62 00 61 00 62 00 79 00 73 00 63 00 6F 00 74 00...
9	REG_BIN	31 00 37 00 37 00 2E 00 6A 00 70 00 67 00 00 00...

Figure 4.4.23: Modification time RecentDocs

Hex Text Application File Metadata OS Account Data Artifacts Analysis Results Context Annotations Other Occurrences

File Metadata

Metadata

Name: .jpg

Number of subkeys: 0

Number of values: 11

Modification Time: 2005-02-03 15:06:42 GMT+00:00

Name	Type	Value
a	REG_SZ	C:\Documents and Settings\johndoe\My Document...
MRUList	REG_SZ	bajihgfd
b	REG_SZ	C:\Documents and Settings\johndoe\My Document...
c	REG_SZ	C:\Documents and Settings\johndoe\My Document...
d	REG_SZ	C:\Documents and Settings\johndoe\My Document...
e	REG_SZ	C:\Documents and Settings\johndoe\My Document...
f	REG_SZ	C:\Documents and Settings\johndoe\My Document...
g	REG_SZ	C:\Documents and Settings\johndoe\My Document...
h	REG_SZ	C:\Documents and Settings\johndoe\My Document...
i	REG_SZ	C:\Documents and Settings\johndoe\Desktop\bir...
j	REG_SZ	C:\Documents and Settings\johndoe\My Document...

Figure 4.4.24: Modification time OpenSaveMRU

Node	Last mod. t.	Key	Type	Value
.jpg	2005/02/0	0	REG_BINARY	6e 00 65 00 73 00 74 00 62 00 6f 00 78 00 74 00 69 00 70 00 73 00 2e 00 74 00 78 00 74 00 00 00
.pdf	2005/02/0	1	REG_BINARY	63 00 6f 00 6f 00 6b 00 69 00 65 00 73 00 2e 00 74 00 78 00 74 00 00 00 54 00 32 00 00 00 00 00
.ram	2005/02/0	MRUListEx	REG_BINARY	01 00 00 00 00 00 00 00 ff ff ff ff
.txt	2005/02/0			
.wav	2005/02/0			
.zip	2005/02/0			
Folder	2005/02/0			
RunMRU	2005/02/0			
Shell Folders	2005/02/0			
ShellImageView	2005/02/0			
StartPage	2005/02/0			
StreamMRU	2005/02/0			

Figure 4.5.0: MRUListEx text document recent order

Node	Last mod. t.	Key	Type	Value
.jpg	2005/02/0	0	REG_BINARY	6e 00 65 00 73 00 74 00 62 00 6f 00 78 00 74 00 69 00 70 00 73 00 2e 00 74 00 78 00 74 00 00 00
.pdf	2005/02/0	1	REG_BINARY	63 00 6f 00 6f 00 6b 00 69 00 65 00 73 00 2e 00 74 00 78 00 74 00 00 00 54 00 32 00 00 00 00 00
.ram	2005/02/0	MRUListEx	REG_BINARY	01 00 00 00 00 00 00 00 ff ff ff ff
.txt	2005/02/0			
.wav	2005/02/0			
.zip	2005/02/0			
Folder	2005/02/0			
RunMRU	2005/02/0			
Shell Folders	2005/02/0			
ShellImageView	2005/02/0			
StartPage	2005/02/0			
StreamMRU	2005/02/0			
Streams	2005/02/0			
StuckRects2	2005/01/2			
tips	2005/01/2			
TrayNotify	2005/02/0			
User Shell Folders	2005/01/2			

Figure 4.5.1: File name “Cookies.txt”

Node	Last mod. t.	Key	Type	Value
.jpg	2005/02/0	0	REG_BINARY	6e 00 65 00 73 00 74 00 62 00 6f 00 78 00 74 00 69 00 70 00 73 00 2e 00 74 00 78 00 74 00 00 00
.pdf	2005/02/0	1	REG_BINARY	63 00 6f 00 6f 00 6b 00 69 00 65 00 73 00 2e 00 74 00 78 00 74 00 00 00 54 00 32 00 00 00 00 00
.ram	2005/02/0	MRUListEx	REG_BINARY	01 00 00 00 00 00 00 00 ff ff ff ff
.txt	2005/02/0			
.wav	2005/02/0			
.zip	2005/02/0			
Folder	2005/02/0			
RunMRU	2005/02/0			
Shell Folders	2005/02/0			
ShellImageView	2005/02/0			
StartPage	2005/02/0			
StreamMRU	2005/02/0			
Streams	2005/02/0			
StuckRects2	2005/01/2			
tips	2005/01/2			
TrayNotify	2005/02/0			
User Shell Folders	2005/01/2			
UserAssist	2005/01/2			

Figure 4.5.2: File name “nestboxtips.txt”

Metadata		
Name: .txt		
Number of subkeys: 0		
Number of values: 3		
Modification Time: 2005-02-03 12:19:07 GMT+00:00		
Values		
Name	Type	Value
0	REG_BIN	6E 00 65 00 73 00 74 00 62 00 6F 00 78 00 74 00...
MRUListEx	REG_BIN	01 00 00 00 00 00 00 00 FF FF FF FF
1	REG_BIN	63 00 6F 00 6F 00 6B 00 69 00 65 00 73 00 2E 00...

Figure 4.5.3: cookies.txt Modification Time



Figure 4.5.4: nestboxtips.txt

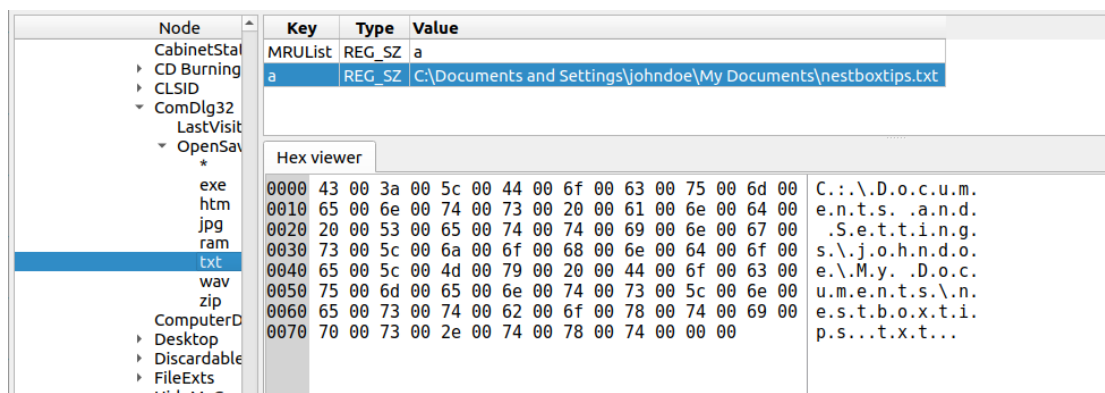


Figure 4.5.6: OpenSaveMRU txt

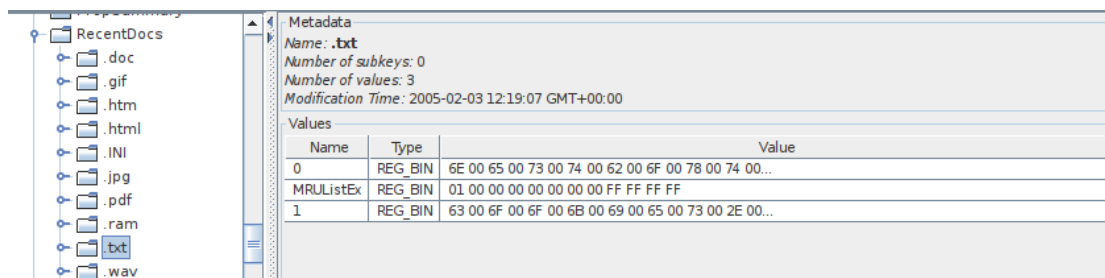


Figure 4.5.7: RecentDocs modified time cookies.txt

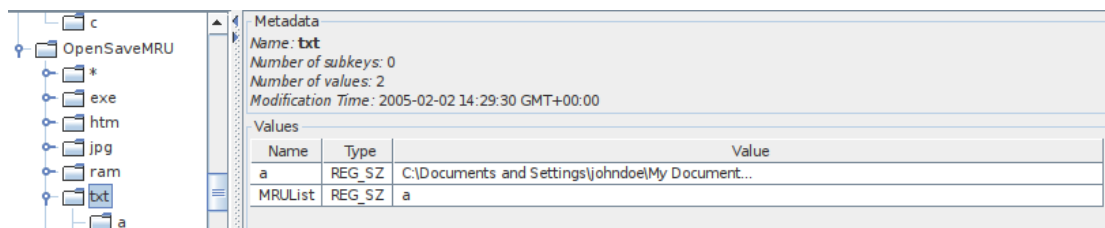


Figure 4.5.8: OpenSaveMRU modified time nestboxtips.txt

Figure 4.6.0: ostbk2b2.htm

Figure 4.6.1: aa010703a.htm

Figure 4.6.2 MRU all htm

Figure 4.6.3 MRU htm Time ostbk2b2.htm RecentDocs

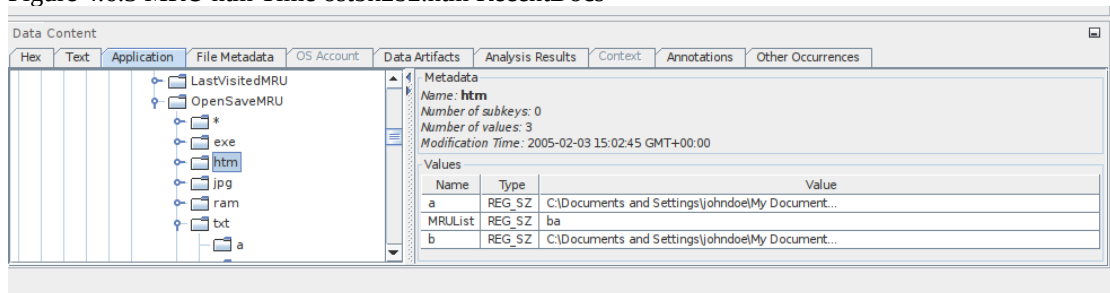


Figure 4.6.4 MRU htm Time aa010703a.htm OpenSaveMRU

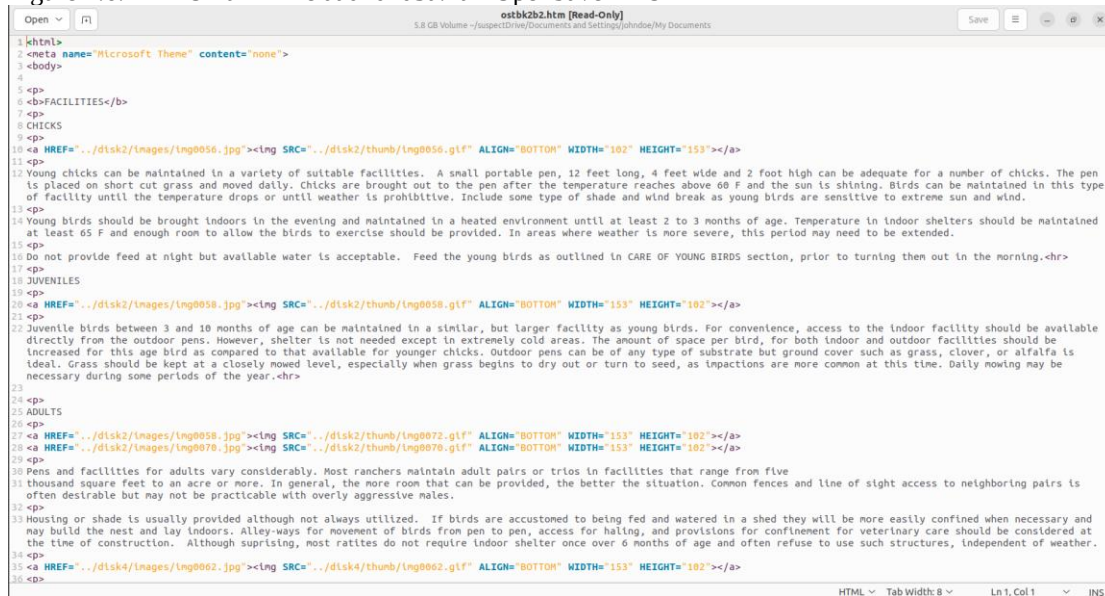


Figure 4.6.5 ostbk2b2.htm inside document



Figure 4.6.6 aa010703a.htm inside links to brds

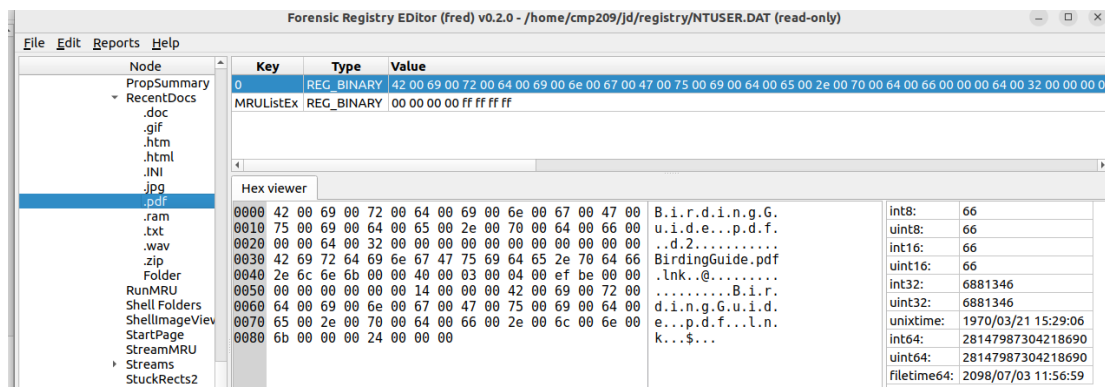


Figure 4.7.0: BirdingGuide.pdf name

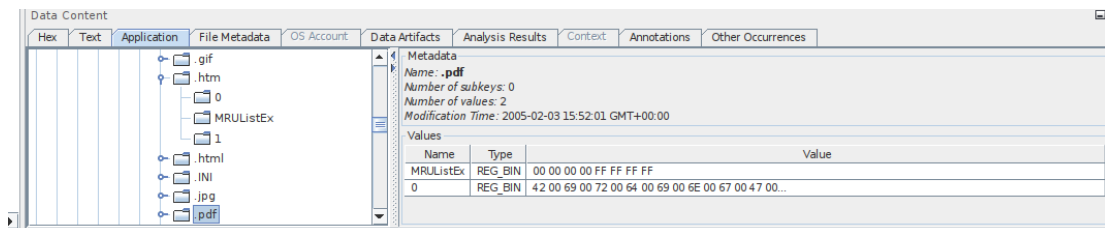


Figure 4.7.1 BirdingGuide.pdf time modified

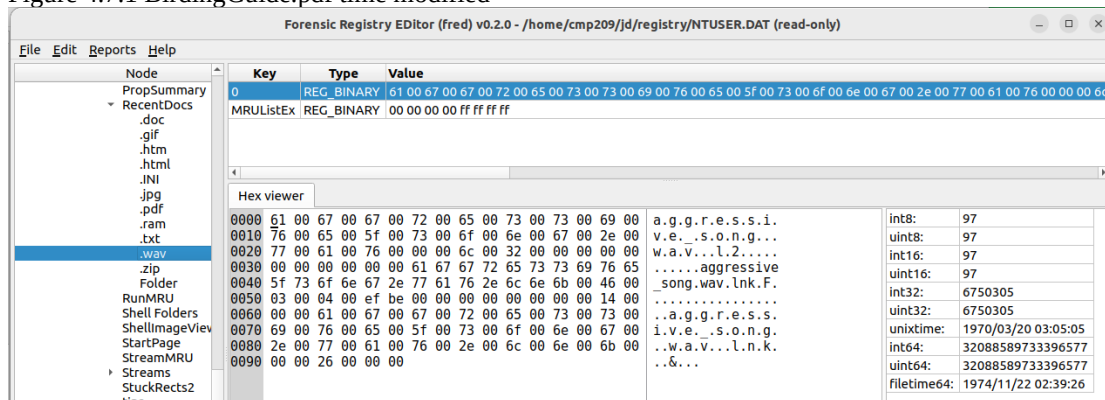


Figure 4.8.0: aggressive_song.wav name

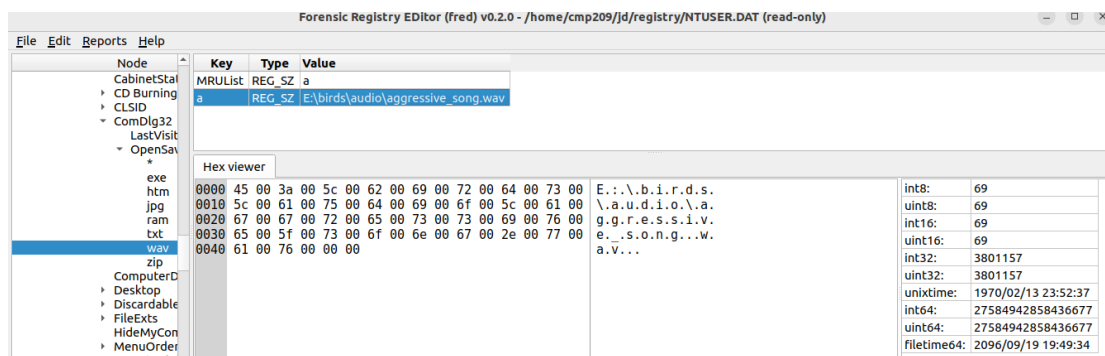


Figure 4.8.1: aggressive_song.wav name

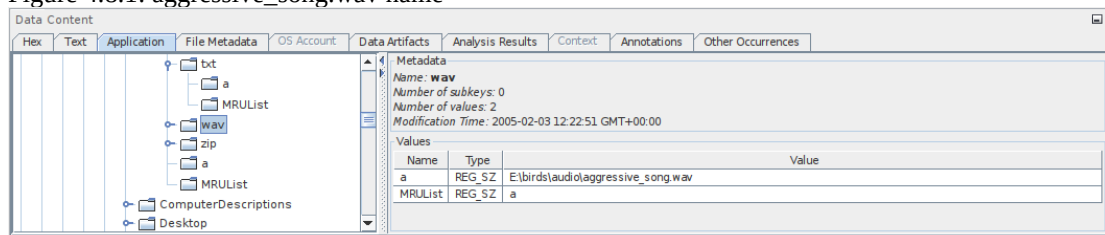


Figure 4.8.2: aggressive_song.wav modified OpenSaveMRU

Appendix C – Network History

Firefox History

First Visit Date	Host Name	Title	URL Length
------------------	-----------	-------	------------

24/01/2005 16:28:20	start.mozilla.org		80
24/01/2005 16:28:20	google.co.uk	Mozilla Firefox Start Page	79
24/01/2005 16:28:37	mozilla.org	About Mozilla	29
24/01/2005 16:28:40	mozilla.org	Mozilla Products	32
24/01/2005 16:28:40	mozilla.org	Thunderbird - Reclaim Your Inbox	44
25/01/2005 11:15:19	office.microsoft.com	Microsoft Office Downloads Home Page	59
25/01/2005 11:15:40	office.microsoft.com	Microsoft Office Online: Internet Explorer version 5.0 or later is required	53
25/01/2005 11:15:46	office.microsoft.com	Microsoft Office Downloads: Office Download Catalog	66
02/02/2005 14:11:46	google.co.uk	Google Search: birds	123
02/02/2005 14:11:54	pbs.org	The Life of Birds	31
02/02/2005 14:12:05	pbs.org	The Life of Birds Songs	47
02/02/2005 14:14:49	amazon.co.uk	Amazon.co.uk: Search Results Books: birds	118
02/02/2005 14:14:59	amazon.co.uk	Amazon.co.uk: Books: Garden Birds Collins Gem S.	107
02/02/2005 14:15:29	amazon.co.uk	Amazon.co.uk: Books: The Secret Lives of Garden Birds	107
02/02/2005 14:15:42	google.co.uk	Google Search: bird wallpaper	153
02/02/2005 14:15:51	naturewallpaper.net	Free Bird Wallpaper - Bald Eagle Albatross Owl Falcon 1024x768	43
02/02/2005 14:15:56	isg10.casalemedia.com	3-Home	44
02/02/2005 14:16:21	naturewallpaper.net	bald_eagle3.jpg	54

02/02/2005 14:18:45	naturewallpaper.net	snow_geese.jpg	55
02/02/2005 14:19:56	uk.shop.com	SHOP.COM - Search results for bald eagle	96
02/02/2005 14:20:19	amos.shop.com	SHOP.COM - Bird Safari - Bird Watcher's Paradise from Choices Direct - Movies is available in Documentary	149
02/02/2005 14:22:25	google.co.uk	Google Search: bird stories	121
02/02/2005 14:22:38	birding.about.com	Bird Stories and Tales	36
02/02/2005 14:22:38	z.about.com	Untitled Document	33
02/02/2005 14:22:45	z.about.com	Current Headlines	39
02/02/2005 14:22:45	z.about.com	Audience Match Data Agent	40
02/02/2005 14:23:04	birding.about.com	Wings of Terror - avian humor article for people interesting in birdwatching and others interested in birds	52
02/02/2005 14:23:04	z.about.com	Related Articles	39
02/02/2005 14:23:56	birding.about.com	Alphabetical Index of Birds	54
02/02/2005 14:24:18	birding.about.com	Specs for Building Birdhouses and for use in Bird House Plans	49
02/02/2005 14:24:32	birding.about.com	How to Build Bird Houses and Nest Boxes	40
02/02/2005 14:24:42	birding.about.com	Free Feeder and Birdhouse Plans Index	65
02/02/2005 14:24:55	birding.about.com	Birding and Birdwatching - Build a Bluebird Nest Box for Wild Birds	53
02/02/2005 14:26:43	birding.about.com	Birding and Birdwatching - Painting or Staining Bird Houses and Feeders	55
02/02/2005 14:27:07	haiths.com	Haith's Wild Bird Foods & Bill Oddie's tips and advice also bird feeder, peanut, birdseed, bird, garden birds, wildlife, nest box, bird	22

		table, birdcare, binocular, bird bath, squirrel, robin,	
02/02/2005 14:28:31	haiths.com	Haith's Wild Bird Foods & Bill Oddie's tips and advice also bird feeder, peanut, birdseed, bird, garden birds, wildlife, nest box, bird table, birdcare, binocular, bird bath, squirrel, robin,	118
02/02/2005 14:40:24	imdb.com	The Birds 1963	36
02/02/2005 14:40:46	imdb.com	Trailers for The Birds 1963	44
02/02/2005 14:41:00	videodetective.com	VideoDetective.com - Movie Trailers, Entertainment Previews, Streaming Video Samples.	51
02/02/2005 14:42:00	videodetective.com	Video Detective Player	191
02/02/2005 14:42:09	videodetective.com	Videodetective.com - Player	105
02/02/2005 14:44:01	news.bbc.co.uk	BBC NEWS News Front Page	22
02/02/2005 14:44:12	news.bbc.co.uk	BBC NEWS Politics Blair defends house arrest plans	50
02/02/2005 14:45:28	bbc.co.uk	BBC Media Selector	139
02/02/2005 14:45:37	bbc.co.uk	BBC Media Selector	260
02/02/2005 14:45:37	news.bbc.co.uk	BBC News Player	154
02/02/2005 14:45:37	news.bbc.co.uk	Untitled	149
02/02/2005 14:45:37	news.bbc.co.uk	BBC NEWS UK Blair defends house arrest plans	95
02/02/2005 15:11:19	news.bbc.co.uk	BBC NEWS Scotland	47
02/02/2005 15:57:40	google.co.uk	Google Search: windows gnupg	152

02/02/2005 15:57:51	gnupg.org	Download - GnuPG.org	30
02/02/2005 15:58:23	winpt.org	WinPT	21
02/02/2005 15:58:48	google.co.uk	Google Search: windows gnupg	118
02/02/2005 15:58:58	openpgp.vie- privee.org	GnuPG Windows	41
02/02/2005 16:52:06	adobe.com	Adobe Reader - Download	52
02/02/2005 16:52:50	adobe.com	Adobe Reader - Download Thank you	194
03/02/2005 12:21:40		Google Search: bird mating calls	135
03/02/2005 12:21:56	whyfiles.org	Chickadee Karaoke	42
03/02/2005 14:59:29		Google Image Search	95
03/02/2005 14:59:40	images.google. co.uk	Google Search: young chicks	127
03/02/2005 14:59:56		Google Image Result for http://freespace.virgin.net/cobber.budgies/images/babyscot_vyoung.jpg	369
03/02/2005 14:59:56	freespace.virgi n.net	Scottie's Baby Pictures - 1	65
03/02/2005 15:01:02		Google Image Result for http://www.insaneanimals.com/items/177.jpg	341
03/02/2005 15:01:08	insaneanimals. com	Young Chicks - Funny Animals, Pets, Cats and Dogs Pictures - Insane Animals	61
03/02/2005 15:02:14		Google Image Result for http://www.cvm.okstate.edu/instruction/kocan/disk2/images/img0056.jpg	383
03/02/2005 15:03:54		Google Image Result for http://people.cornell.edu/pages/sah67/chicks2.jpg	346

03/02/2005 15:03:54	people.cornell.edu	Eco Gallery '04	49
09/02/2005 11:27:00		Google Search: bird screensavers	135
09/02/2005 11:27:12	traveltex.com	Screensavers	56

Appendix D – Timeline

Images taken with **Canon PowerShot SD100** Camera (with capture date and time)

13/06/2004

18:28 - Photo taken: 01243351.jpg.
 19:44 - Photo taken: 02791503.jpg.
 20:39 - Photo taken: 03185759.jpg.
 21:34 - Photo taken: 05475951.jpg.

17/06/2004

18:42 - Photo taken: 02963839.jpg.
 18:42 - Photo taken: 03184607.jpg.
 18:43 - Photo taken: 02962639.jpg.

18/06/2004

23:31 - Photo taken: 03074343.jpg.
 23:31 - Photo taken: 03180791.jpg.
 23:32 - Photo taken: 03181927.jpg.
 23:33 - Photo taken: 02903551.jpg.
 23:49 - Photo taken: 03538975.jpg.

19/06/2004

19:13 - Photo taken: 03241879.jpg.
 19:49 - Photo taken: 03188831.jpg.

20/06/2004

18:01 - Photo taken: 05069311.jpg.

22/06/2004

18:56 - Photo taken: 03112503.jpg.
 21:10 - Photo taken: 03186407.jpg.

27/06/2004

18:28 - Photo taken: 02792151.jpg.
 18:28 - Photo taken: 02792407.jpg.

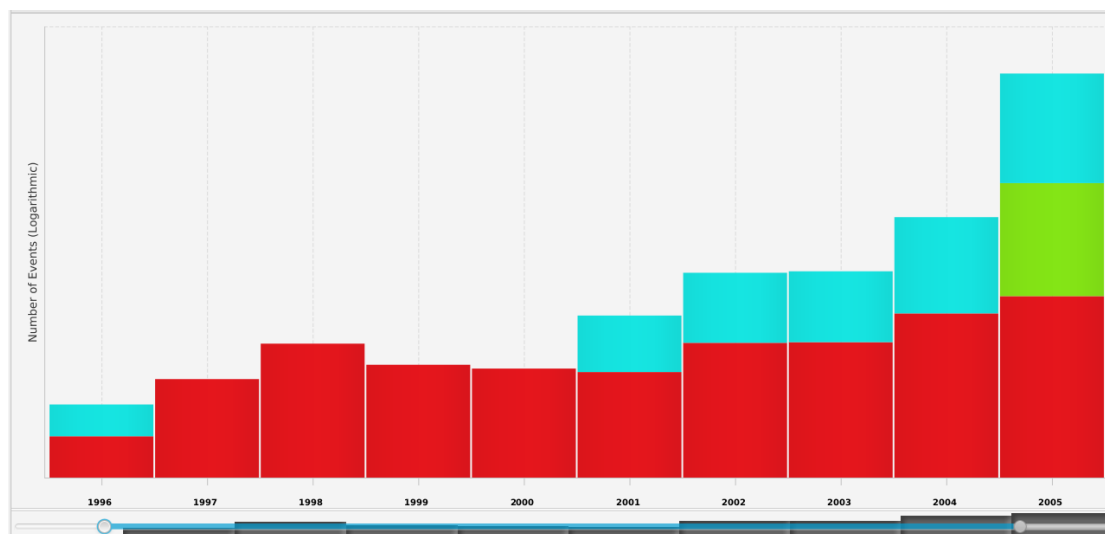


Figure 5.1: Autopsy Timeline of Events

Firefox History Timeline

24/01/2005

16:28:40 – Mozilla Thunderbird was downloaded.

02/02/2005

14:11:46 – Google search of “birds” was performed.

14:11:54 – pbs.org “The Life of Birds by David Attenborough” was visited.

14:12:05 – pbs.org article “The Life of Birds | Songs” was visited.

14:14:49 – Amazon.com books search of “birds” was performed.

14:14:59 – Amazon.com book “Garden Birds Collins Gem S.” was viewed.

14:15:29 – Amazon.com book “The Secret Lives of Garden Birds” was viewed.

14:15:42 – Google search of “bird wallpaper” was performed.

14:15:51 - Naturewallpaper.net “Free Bird Wallpaper - Bald Eagle Albatross Owl Falcon 1024x768” was visited.

14:16:21 - naturewallpaper.net “bald_eagle3.jpg” was visited.

14:18:45 - naturewallpaper.net “snow_geese.jpg” was visited.

14:19:56 - shop.com search of “bald eagle” was performed.

14:20:19 - shop.com “Bird Safari - Bird Watcher's Paradise from Choices Direct” was visited.

14:22:25 – Google search of “bird stories” was performed.

14:22:38 - birding.about.com “Bird Stories and Tales” was visited.

14:23:04 – birding.about.com “Wings of Terror - avian humor article for people interesting in birdwatching and others interested in birds” was visited.

14:23:04 – birding.about.com related articles page (referring to previous timestamp) was visited.

14:23:56 - birding.about.com “Alphabetical Index of Birds” was visited.

14:24:18 - birding.about.com “Specs for Building Birdhouses and for use in Bird House Plans” was visited.

14:24:42 - birding.about.com “Free Feeder and Birdhouse Plans Index” was visited.

14:24:55 - birding.about.com “Birding and Birdwatching - Build a Bluebird Nest Box for Wild Birds” was visited.

14:26:43 - birding.about.com “Birding and Birdwatching - Painting or Staining Bird Houses and Feeders” was visited.

14:27:07 - haiths.com “Haith's Wild Bird Foods & Bill Oddie's tips and advice also bird feeder, peanut, birdseed, bird, garden birds, wildlife, nest box, bird table, birdcare, binocular, bird bath, squirrel, robin,” was visited.

14:40:24 – imdb.com “The Birds 1963” was visited.

14:40:46 – imdb.com “Trailers for The Birds 1963” was visited.

15:57:40 – Google search of “windows gnupg” was performed.

15:57:51 – GnuPG was installed.

16:52:06 – Adobe Reader was installed.

03/02/2005

12:21:40 – Google search of “bird mating calls” was performed.

12:21:56 - whyfiles.org "Chickadee Karaoke" was visited.
14:59:40 – Google search of "young chicks" was performed.
14:59:56 – freespace.virgin.net "babyscot_vyoung.jpg" was visited.
14:59:56 - freespace.virgin.net "Scottie's Baby Pictures - 1" was visited.
15:01:02 - insaneanimals.com "177.jpg" was visited.
15:01:08 - insaneanimals.com "Young Chicks - Funny Animals, Pets, Cats and Dogs Pictures - Insane Animals" was visited.
15:02:14 - cvm.okstate.edu "img0056.jpg" was visited.
15:03:54 - people.cornell.edu "chicks2.jpg" was visited.
15:03:54 - people.cornell.edu "Eco Gallery '04" was visited.
09/02/2005
11:27:00 – Google search of "bird screensavers" was performed.
11:27:12 - traveltex.com "Screensavers" was visited.

Figure 5.2: Simplified Firefox History Timeline